

Wildlife Hazard Assessment and Management Plan

**Naval Air Station Sigonella,
Sicily, Italy**

April 2022- April 2023



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CNREURFCENT



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LIST OF ACRONYMS

AC	Advisory Circular
AOA	Airfield Operations Area
AFSAS	Air Force Safety Automated System
AFSC	Air Force Safety Center
AMC	Air Mobility Command
AOPS	Air Operations Officer
APHIS	Animal Plant Health Inspection Service
ASO	Aviation Safety Officer
ATC	Air Traffic Control
ATIS	Automated Terminal Information Systems
AVOC	Airfield Vehicle Operators Course
BASH	Bird/Wildlife Aircraft Strike Hazard
BMZ	BASH Mitigation Zones
BWC	Bird Watch Condition
BWG	Bird Working Group
CFR	Code of Federal Regulation
CNREURAFCENT	Commander, Navy Region Europe, Africa, Central
CONUS	Continental United States
CNIC	Commander Naval Installation Command
CMA	Controlled Movement Area
DOD	Department of Defense
DRMO	Defense Reutilization Marketing Office
ERS	Expeditionary Reconnaissance Squadron
EU	European Union
FAA	Federal Aviation Administration
FGS	Final Governing Standards
FOD	Foreign Object Debris
FLIR	Forward Looking InfraRed
FY	Fiscal Year
HN	Host Nation
IAW	In Accordance With
ITAF	Italian Air Force
MBTA	Migratory Bird Treaty Act
NASSIG	Naval Air Station Sigonella
NATO	North Atlantic Treaty Organization
NAVFAC	Naval Facilities Engineering Systems Command
NOTAM	Notice to Airmen
NRM	Natural Resources Manager
NRMP	Natural Resource Management Plan
OEBGD	Overseas Environmental Baseline Guidance Document



LIST OF ACRONYMS (CONTINUED)

OCONUS	Outside the Continental U.S.
OPSO	Operations Officer
PSA	Primary Surface Area
PW	Public Works (<i>Navy Grounds Maintenance/Construction Services</i>)
PWO	Public Works Officer
RHDV2	Rabbit Hemorrhagic Disease 2
RMI	Risk Management Information
RS	Reconnaissance Squadron
RSL	Ready Service Locker
RWY	Runway
SOP	Standard Operating Procedure
SV	Site Visit
US	United States (of America)
USAF	United States Air Force
USDA	United States Department of Agriculture
USMC	United States Marine Corps
USN	United States Navy
WASH	Bird/Wildlife Avoidance Strike Hazard
WESS	Web Enabled Safety System
WHA	Wildlife Hazard Assessment
WHMP	Wildlife Hazard Management Plan
WS	Wildlife Services



EXECUTIVE SUMMARY

The United States Department of Agriculture (USDA), Animal and Plant Health Inspection Service (APHIS), Wildlife Services (WS), in cooperation with Naval Air Station Sigonella (NASSIG), developed this Wildlife Hazard Assessment (WHA) to provide a formalized report of systematic data regarding Bird/Wildlife Aircraft Strike Hazard(s) (BASH) to aircraft and human health and safety at NASSIG.

A review and evaluation of bird strike records during the WHA period at NASSIG was completed to assess BASH risks. Strike data from Fiscal Year (FY) 2012 through 2023 was included. Current management practices in relation to BASH at NASSIG were reviewed and included a characterization of wildlife attracting habitats both on and off base. As an Outside the Continental United States (OCONUS) base, NASSIG does not fall under the same laws as a Continental United States (CONUS) airfield. An overview of laws regulating BASH activities is provided in section 4.0.

Standardized point count surveys and driving observations, were used to identify and document hazardous wildlife activity at NASSIG. Sixty-one species of animals (fifty-six species of birds and five species of mammals) were observed at NASSIG during the APR2022–APR2023 WHA period. Blackbirds, columbids, and small songbirds were the top three most abundant bird guilds observed at NASSIG during airfield surveys, respectively. Mammals, European rabbits (*Oryctolagus cuniculus*) in particular, were observed in high densities year-round across the entire airfield. Wildlife Hazard Assessment airfield survey data were examined to identify species trends on the airfield at NAS Sigonella and the potential BASH issues that they pose.

USDA-WS has several recommendations to build the structure of and ensure the continued success of the BASH program at NASSIG via the reduction of BASH hazards and attractions on and near the airfield and the collection of needed data to track BASH risk. These include but are not limited to: proper bird strike reporting procedures, addressing drainage and flooding issues, vegetation management, and the removal of unnecessary structures near the airfield.

1.0 INTRODUCTION

The United States Navy (USN) requires naval air bases without USDA-WS biologists to complete WHAs every 10 years as stated in CNIC Instruction 3750.1B. The last WHA performed at NASSIG was completed in September 2017 by Commander, Navy Installations Command (CNIC)/ Naval Facilities Engineering Systems Command (NAVFAC) BASH Natural Resources Program Manager. Several types of bi-monthly surveys were conducted at NASSIG from 15APR2022 through 15APR2023, providing the material for the WHA. NASSIG was fully staffed by a USDA-WS Wildlife Biologist during this period. All surveys were conducted by USDA-WS biologists. General observations of BASH conditions (outside of standardized survey times) were also collected during this period.

1.1 Overview of wildlife hazards at airports

Bird strikes to aircraft are a serious safety and economic problem. In 2022 it is estimated wildlife strikes caused \$385 million dollars of damage and almost 68,000 hours of downtime, although these estimates may be on the high end (Dolbeer et al. 2023). Military aircraft are more vulnerable to bird strikes than commercial aircraft because they are designed more for speed and agility than safety. For example, many fighters have only one engine, increasing the probability of crashing if the engine ingests a bird and shuts down. Commercial jets can still fly with the loss of an engine because they generally have more than one



(Conover 2001). Military aircraft also fly very close to the ground compared to other aircraft; this fly zone is also shared by most birds. Navy and United States Marine Corp (USMC)-wide, in FY21, there were a reported 1,251 bird/wildlife strikes. The total estimated cost of all strikes was \$60,843,306 (M. J. Greene, U.S. Navy, personal communication).

The Air Force Safety Center (AFSC) maintains the most accurate records of bird hazards to military aircraft. In FY21, the AFSC recorded 3,632 total BASH related events within the United States Air Force (USAF) that totaled approximately \$18.4 million in damages. There were no Class A mishaps, three Class B mishaps, sixty-six Class C, 139 Class D, 231 Class E, and 3,293 hazards (no cost) (D. P. Sullivan, U.S. Air Force Bird Aircraft Strike Hazard Team, personal communication).

Military regulations are based on the Federal Aviation Administration (FAA) Advisory Circulars (AC). The FAA defines a wildlife strike to an aircraft in AC 150/5200-32B. The definition is as follows:

A wildlife strike has occurred when:

1. A strike between wildlife and aircraft has been witnessed.
2. Evidence or damage from a strike has been identified on an aircraft.
3. Bird or other wildlife remains, whether in whole or in part, are found:
 - (a) Within 250 feet of a runway centerline or within 1,000 feet of a runway end unless another reason for the animal's death is identified or suspected.
 - (b) On a taxiway or anywhere else on or off the airport that you have reason to believe was the result of a strike with an aircraft. Examples might be:
 - (i) A bird found in pieces from a prop strike on a taxiway.
 - (ii) A carcass retrieved within 1 mile of an airport on the final approach or departure path after someone reported the bird falling out of the sky and a report of a probable wildlife strike.
4. The presence of birds or other wildlife on or off the airport had a significant negative effect on a flight (i.e., aborted takeoff, aborted landing, high-speed emergency stop, or the aircraft left pavement area to avoid collision with wildlife) (FAA 2013).

The Code of Federal Regulation (CFR), Title 14, Part 139.137 requires certificated airports to conduct a WHA when any of the following events occurs on or near the airport:

1. An air carrier aircraft experiences multiple wildlife strikes.
2. An air carrier aircraft experiences substantial damage from striking wildlife.
3. An air carrier aircraft experiences an engine ingestion of wildlife.
4. Wildlife of a size, or in numbers, capable of causing an event described in paragraph (b)(1), (2), or (3) of 14 CFR § 139.337 is observed to have access to any airport flight pattern or aircraft movement area (FAA 2018).

It is required by CNICINST 3750.1B, that a WHA be conducted every 10 years unless continual monitoring has occurred. While items 1, 3, and 4 above have occurred at NASSIG since the last WHA was completed in 2017, this WHA is being completed to satisfy the 10-year requirement by CNIC at certificated airports. NASSIG is subject to very high bird activity on and around the airfield due to the ample amount of habitat in the area.



1.2 Authority of Wildlife Services

The USDA-APHIS-WS had a working agreement with Commander, Navy Region Europe, Africa, Central (CNREURAFCENT) (APHIS Agreement #: 34-WT-20-7100-0451-IA) for the mitigation of potential and realized wildlife hazards at NASSIG and other Region airfields. This five-year agreement was in effect during the WHA data collection period. This agreement has since been rolled into the national CNIC Agreement (APHIS Agreement #: 34-WT-24-7100-0323-IA). The CNIC agreement states the following:

“The USDA is authorized to protect American agriculture and other resources from damage associated with wildlife. The primary authority for WS is the Act of March 2, 1931, (46 Stat. 1468; 7 U.S.C.426-426b) as amended, and the Act of December 22, 1987, (101Stat. 1329-331, 7 U.S.C. 426c). Wildlife Services activities are conducted in cooperation with other Federal, State and local agencies; private organizations; and individuals. Department/agency cooperation between the USDA and DOD also is based on language contained in the Sikes Act (16 USC 670a-670o, 74 Stat. 1052), Department of Defense (DOD) Instructions 4150.07 (DOD Pest Management Program) and 4715.03 (Natural Resources Conservation Program), and the memorandum of understanding (# 12-34-71-0007-MU) between the DOD and the USDA, APHIS, Wildlife Services”.

2.0 OBJECTIVES

This WHA shall complete the following objectives:

1. Analysis of the circumstances that prompted the assessment.
2. Identification of the wildlife species observed during the WHA period and their numbers, locations, local movements, and seasonal occurrences.
3. Identification and location of features on and near the airport that attract wildlife.
4. A description of wildlife hazards to air operations.
5. Recommended actions for reducing wildlife hazards to air operations.

3.0 AIRPORT/INSTALLATION OVERVIEW

3.1 Site Description

NASSIG is a tenant of the Italian Air Force (ITAF), who have military and administrative control of ITAF Base Sigonella. USN operates on the western side of the base, while ITAF and North Atlantic Treaty Organization (NATO) mainly operate on the eastern side. In this WHA, all references to NASSIG will include both USN and ITAF/NATO. NASSIG is a 2,180-acre naval air station located in the commune of Lentini, Sicily, Italy (Figure 1). The air station has several natural characteristics that make most of the site appealing to birds. NASSIG is bordered by large agricultural fields and pastures on all sides. The Dittaino River runs along the northern border of the base and the Gornalunga River runs along the southern border. The rivers bordering NASSIG attract a variety of bird species that can pose significant threats to aircraft. There are also large areas of vegetation and grasslands interspersed with ephemeral and permanent water sources (e.g., seasonally flooded wetlands and constant standing water in drainages) that attract multiple species of birds and mammals. NASSIG has two parallel runways running East/West: Runway (RWY) 10R/28L and

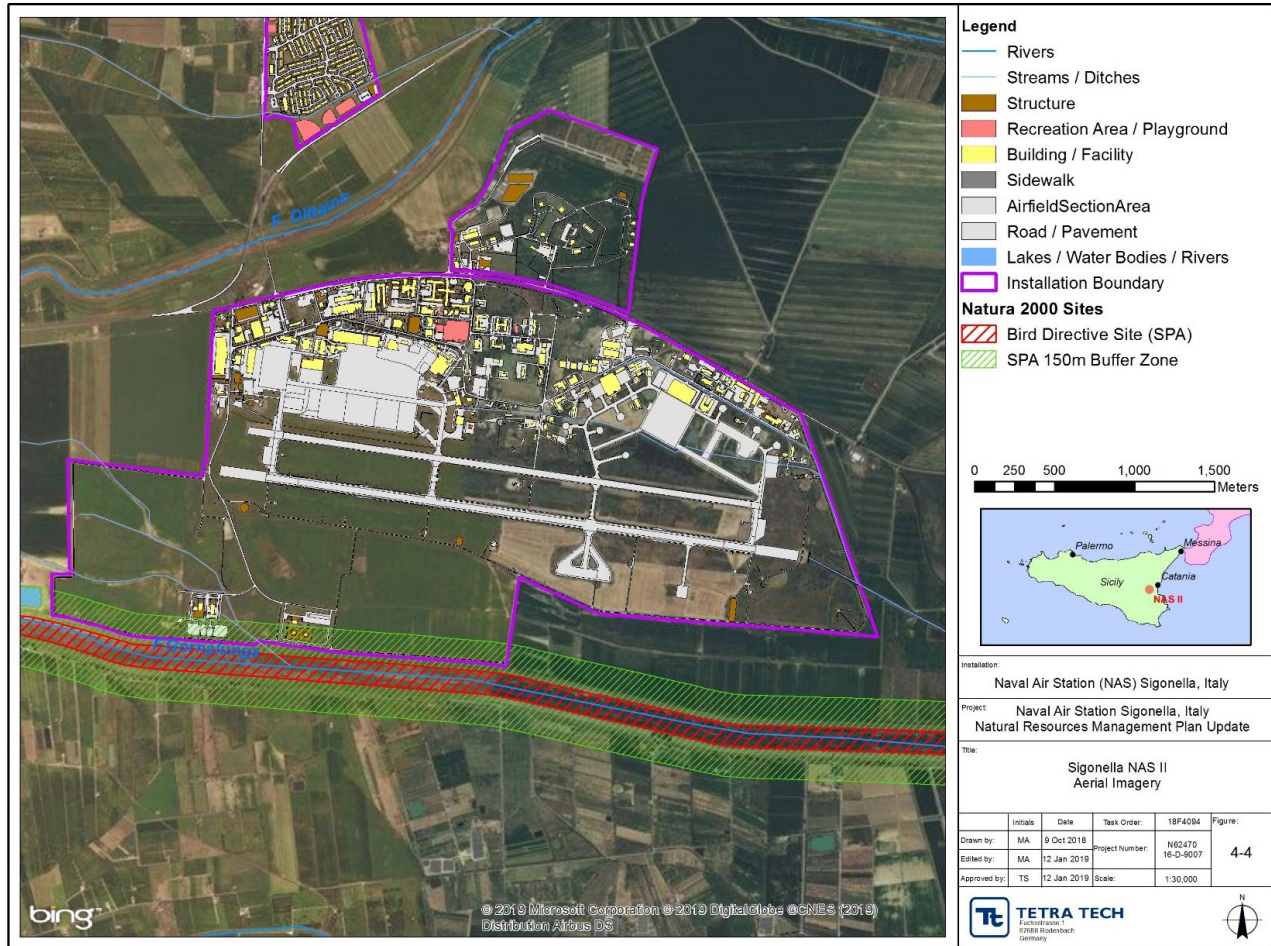
RWY 10L/28R. 10L/28R is not currently used as a runway; it is only used for taxiing to/from the active runway. An ongoing NATO construction project is widening 10L/28R and extending the length of both “runways” from just exceeding 8,000 feet to approximately 11,000 feet. This will allow both runways to be utilized. There are both rotary-wing aircraft and fixed-wing aircraft stationed at NASSIG.

Figure 1. NASSIG location.





Figure 2. NASSIG (NAS2) Installation Layout (NASSIG NRMP, 2019)





3.2 Airport Operations and Facilities Information

NASSIG hosts a wide variety of USN, USAF, ITAF, and NATO aircraft. Two C-26D “Metroliners” are stationed at NASSIG to transport important cargo and passengers across Europe. Detachments of P-8As, C-40As, C-130s and MH-60Ss also support the USN mission at NASSIG. C-2As support USN carrier strike groups operating in the Mediterranean. USAF 7th Reconnaissance Squadron (RS) operates RQ-4 drones. USAF 324th Expeditionary Reconnaissance Squadron (ERS) operated MQ-9s during the period of the WHA but have since been relocated. USAF Air Mobility Command (AMC) operates the Patriot Express/Rotator and passenger terminal along with a cargo terminal. ITAF has P-72s stationed at NASSIG. NATO MQ-4s are also stationed at NASSIG.

This is not a complete list of operational aircraft at NASSIG; there are several other types of aircraft that regularly transit through and/or train at the installation.

During the WHA data collection period (APR22-APR23), there were a total of 14,312 air operations at NASSIG that included arrivals (5,919), departures (5,998), and overflights (2,395) (B. R. Moon, personal communication).

The NASSIG Bird Working Group (BWG) meets quarterly to share pertinent information that can be used in the recommendation of actions and changes to operations in order to reduce hazards on the airfield. The BWG consists of representatives that are involved in airfield operations and management from the following departments (NASSIGINST 3750.3F):

- 1) NASSIG Operations Officer (OPSO) (Chair)
- 2) NASSIG Aviation Safety Officer (ASO)
- 3) NASSIG Air Traffic Control (ATC) Officer
- 4) NASSIG Airfield Manager
- 5) ASOs for all tenant aviation units
- 6) ITAF ASO
- 7) NASSIG Public Works Officer (PWO)
- 8) Environmental Officer/Natural Resources Manager (NRM)
- 9) USDA-WS Biologist

To support the BWG, USN has an Inter-agency Agreement (34-WT-20-7100-451-IA) with USDA-WS to assist in the mitigation of wildlife strike hazards to aircraft operating on the airfields at NASSIG. The intent of this agreement is to reduce the number of birds that fly, nest, roost, or loaf in areas around the airfield on NASSIG as they can present a strike hazard to aircraft. Additionally, all other vertebrates posing a safety risk are to be considered for management (e.g., mammals, reptiles, fish).

3.3 Wildlife Strike History

Bird strikes at NASSIG, primarily those that caused damage, have been reported since at least 1981. In FY20, CNIC, including NASSIG shifted from use of the Web Enabled Safety Systems (WESS) to the Risk Management Information (RMI) system; data from before this transition were transferred into RMI, but are incomplete and important details may be missing. In order to help find missing data points, the “One-liner” and “Event Narrative” of the applicable report were referenced. For this WHA, the strikes from FY12-APR23 (267 strike events) will be analyzed with the focus on strike events during the WHA period that are listed in Table 1. The data include strikes from RMI (USN & USMC), AFSAS (USAF), and ITAF/NATO. Bird strike data reports with ITAF/NATO are only available beginning in FY18; previous data is unavailable.



Wildlife Hazard Assessment NASSIG



Table 1. Bird/Wildlife Strikes at NASSIG APR22-APR23.

DATE	TIME	AIRCRAFT	DAMAGE/ CLASS	SPECIES 1	SPECIES 2
2-Apr-22	30	P-8A	N/A	Common Whitethroat	
13-Apr-22	1000	P-8A	N/A	Black-headed Gull	
18-Apr-22	1120	P-8A	N/A	Swift species	Bank Swallow X2
19-Apr-22	1900	P-8A	N/A	Swift species	Bank Swallow X2
22-Apr-22	30	P-8A	N/A	Swift species	
25-Apr-22	1655	P-8A	\$913,100/ B	White Stork	
29-Apr-22	2055	P-8A	N/A	Swift species	Bank Swallow X2
29-Apr-22	UNK	P-8A	N/A	Swift species X2	
30-Apr-22	UNK	RQ-4	N/A	European Starling	
2-May-22	UNK	RQ-4	N/A	Spotted Flycatcher	
3-May-22	1700	P-8A	N/A	Common Swift	
13-May-22	1430	P-8A	N/A	Common House Martin	
24-May-22	UNK	P-8A	N/A	"Old World Sparrow"	
10-Jun-22	1600	P-8A	N/A	Common Swift	
21-Jun-22	2130	P-8A	N/A	Savi's Pipistrelle	Common Swift X2
22-Jun-22	1230	P-8A	N/A	Barn Swallow	
27-Jun-22	1100	P-8A	N/A	Barn Swallow	
29-Jun-22	1730	P-8A	N/A	Barn Swallow	
3-Jul-22	615	P-8A	N/A	Savi's Pipistrelle X2	Common Swift X2
12-Jul-22	900	P-8A	N/A	"Old World Sparrow" X2	
19-Jul-22	UNK	P-8A	N/A	Barn Swallow X2	
29-Jul-22	2100	C-2A	N/A	Eurasian Kestrel	
30-Jul-22	100	P-8A	N/A	Eurasian Stone-curlew X2	
2-Aug-22	1930	P-8A	N/A	Savi's Pipistrelle	Common Swift X2
17-Aug-22	UNK	P-8A	N/A	European Bee-eater X2	
28-Aug-22	1000	P-8A	N/A	Unknown	
29-Aug-22	1530	P-8A	N/A	Savi's Pipistrelle	
2-Sep-22	630	P-8A	N/A	Spotted Flycatcher	
5-Sep-22	615	P-8A	N/A	Red-backed Shrike	
6-Sep-22	400	P-8A	N/A	"Perching Bird" species	
7-Sep-22	730	P-8A	N/A	Barn Swallow	
10-Sep-22	2010	P-8A	N/A	Willow Warbler	Spotted Flycatcher
12-Sep-22	1000	P-8A	N/A	Barn Swallow	
21-Sep-22	815	P-8A	\$540/ E	Common Nightingale	
28-Sep-22	900	P-8A	N/A	Cattle Egret	
28-Sep-22	UNK	P-8A	N/A	Cattle Egret	
12-Oct-22	1500	P-8A	N/A	Eurasian Kestrel	
16-Oct-22	330	P-8A	N/A	Eurasian Blackcap	
16-Jan-23	1800	P-8A	N/A	European Golden-Plover	



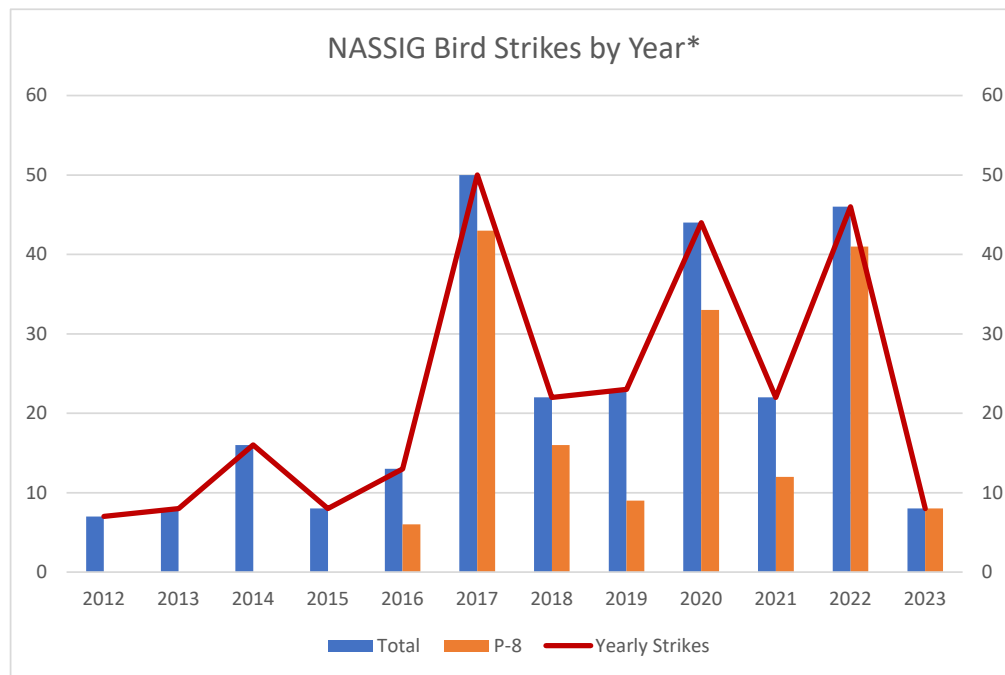
Table 1(cont.) Bird/Wildlife Strikes at NASSIG APR22-APR23

4-Feb-23	520	P-8A	N/A	European Golden-Plover	
8-Feb-23	1100	P-8A	N/A	Common Buzzard X2	
27-Feb-23	1055	P-8A	N/A	Rock Pigeon	
4-Mar-23	1500	P-8A	N/A	White Wagtail	
14-Mar-23	1600	P-8A	N/A	Western Marsh Harrier	
22-Apr-23	1030	P-8A	N/A	Western Marsh Harrier	
30-Apr-23	2330	P-8A	N/A	Tree Pipit	

Between FY12-FY22, there were at least ten damaging bird strikes, causing over \$1.2M in damages, reported at NASSIG. Of these ten damaging strikes, seven of them involved P-8As and cost \$1.16M total in damages.

Between FY12 and FY20, there were a total of 190 bird strikes at NASSIG. In FY21, the year USDA-WS biologist arrived on station, twenty-one strikes were reported. In FY22, forty-six strikes were reported. During FY23 through 30APR23, 10 strikes were reported. This increase in reported strikes is likely due to increased strike reporting and sampling training opportunities rather than a doubling of the hazard present.

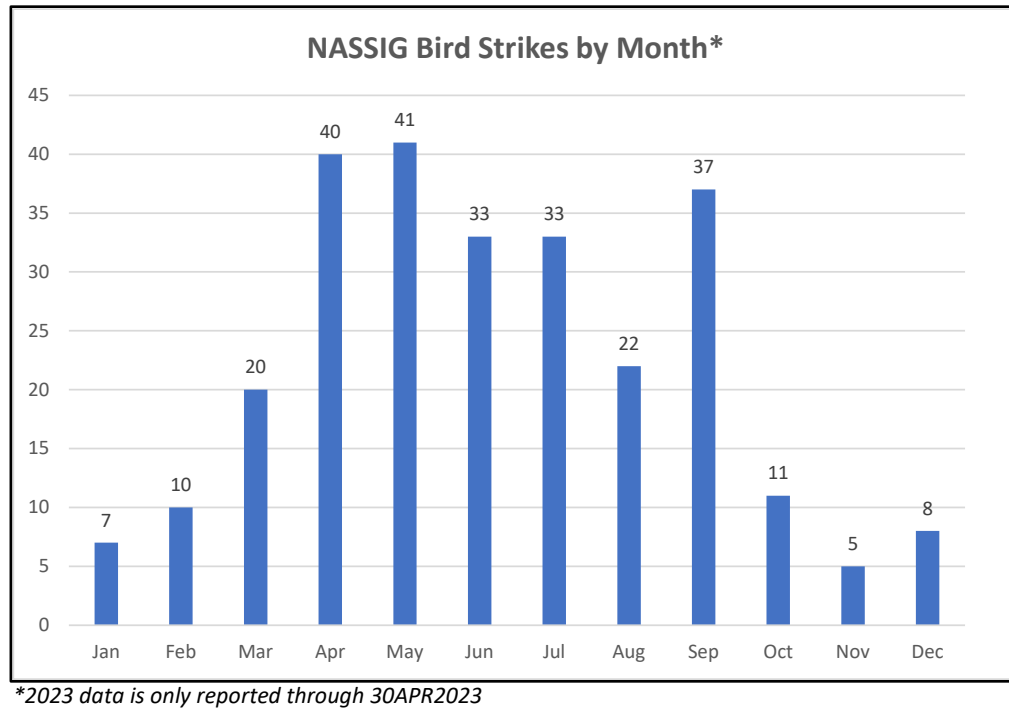
Figure 3. Bird strike events at NASSIG FY12–APR23.



*2023 data is only reported through 30APR2023



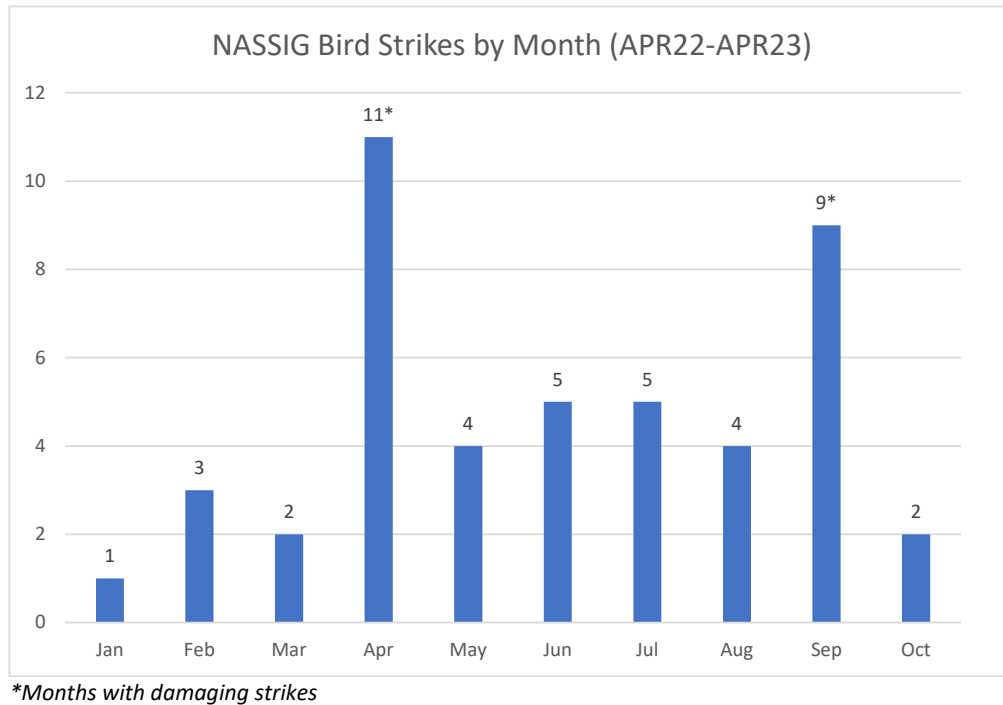
Figure 4. Bird strike events per month at NASSIG during FY12–APR23.



There was an average of 25.4 strike events per year at NASSIG between October 2012 through April 2023 (~10.5 years). On average, one strike event per year caused damage to aircraft. NASSIG experienced sixty-seven strike events during the WHA data collection period (APR22-APR23) (Figure 5). During this time, two strikes were known to have caused at least \$913,640 total in aircraft damage. One strike involved a white stork (\$913,100/Class B) and the second a common nightingale (\$540/Class E). The most frequently struck animals between APR22-APR23 included twenty-eight martins/swallow/swifts, sixteen passerines (e.g. songbird), six raptors, and five bats (Table 1). Some strikes did involve more than one individual animal being struck. There was a total of 14,312 air operations during this period, including arrivals, departures, and overflights (B.J. Moon, personal communication). With forty-six known strike events during this period, there was an average of 32.14 strike events per 10,000 air operations at NASSIG during the WHA data collection period (APR22-APR23).



Figure 5. Total bird strike events per month at NASSIG during WHA.



3.4 Wildlife Strike Reporting

Bird remains are typically found by squadron maintenance personnel through blood smears and feathers (snarge), or actual damage to the aircraft. Generally, remains are placed in a designated freezer in Hangar 630, which USDA-WS checks on a regular basis. All bird strikes are reported by squadron aviation safety officers to the Navy Safety Center through RMI as well as to NASSIG ASO. Appendices 1-2 show the bird strike reporting procedures and forms given to airfield personnel and/or squadron ASOs.

3.5 Current Management Practices

NASSIG and their airfield BASH operations are managed in accordance with the protocols established in several documents, including ITAF AIRPORT COMMAND SIGONELLA POP-SIV-003 BIRD/WILDLIFE AVOIDANCE STRIKE HAZARD (WASH) PLAN, CNICINST 3750.1B NAVY BIRD ANIMAL AIRCRAFT STRIKE HAZARD PROGRAM IMPLEMENTING GUIDANCE(2023), NASSIGINST 3750.3F BIRD AND ANIMAL AIRCRAFT STRIKE HAZARD PLAN(2023), CNIC BASH Manual (2018), APHIS Agreement #: 34-WT-24-7100-0323-IA, NASSIG Natural Resource Management Plan (NRMP) (2019) and USDA Wildlife Hazard Management at Airports Manual (2020). These documents combined form a cohesive plan to ensure the continued safe and efficient operation of the airfield at NASSIG.

ITAF WASH Team is largely responsible for BASH remediation on the airfield at NASSIG and for providing the vast majority of wildlife dispersal. USDA-WS provides additional dispersal assistance with the utilization of a vehicle and pyrotechnics. Current efforts by USDA-WS to reduce bird strikes consist entirely of non-lethal harassment as well as habitat management recommendations to NASSIG Operations.

Habitat manipulation offers the most viable and long-term solution for the reduction of wildlife around airfields. However, it is not always practical nor achievable due to constraints with funding, staffing, and



environmental impact. All habitat management must be coordinated with Host Nation (HN) and is usually also conducted by HN. Harassment is a short-term tool that helps to establish wildlife-free buffer zones in and around the airfield operating environment to further reduce bird and other wildlife strikes.

Wildlife monitoring on and around the airfield is crucial in aiding bird harassment and the reduction of aircraft strikes. Survey trend data are provided to the ASO during BASH WG meetings as a precautionary measure to warn pilots of potential bird activity as it relates to seasonal changes, the time of day, and specific locations. When possible, and when it does not affect the mission, flight times can be adjusted to when risk of a wildlife strike is the lowest. Wildlife monitoring can help to distinguish high and low risk areas, allowing for the concentration of harassment efforts. It can also be used to consider the need for lethal control.

Bird watch conditions (BWC) place airfield conditions into one of four categories (low, moderate, severe, and prohibitive) (NASSIGINST 3750.3F). ITAF Tower personnel set BWC under advisement of multiple entities on the airfield, including USDA-WS and the ITAF WASH team.

USDA-WS biologists will continue to work with all flying squadrons at NASSIG to ensure they have proper strike collection materials and understanding of the standard operating procedure(s) (SOP) of strike reporting.

3.6 Wildlife Attractants

Wildlife attractants are resources that attract the wildlife that use them for various life-sustaining functions; they can be grouped into the following categories: food, water, vegetation (cover), and artificial environment. Off-site attractants will also be discussed and may include food, water, and/or cover occurring off-base near NASSIG.

At NASSIG, there are many wildlife species that either directly or indirectly pose strike threats to aircraft. Various bird species pose the greatest risk as they routinely either pass over or use the airfield and its surrounding areas. European rabbits pose an indirect threat by attracting predatory and scavenging birds and mammals to the airfield. Rabbits in Sicily are also prone to become infected with rabbit hemorrhagic disease 2 (RHDV2), which is nearly always fatal. Transmission rates are likely high due to the high density of rabbits on NASSIG and rabbits are often found dead on the airfield, becoming an attractant. Feral dogs (*Canis familiaris*) also occasionally cross the runways, creating a strike hazard. These feral dogs are also a threat to pedestrians; multiple attacks have been documented.

3.6.1 Food Sources

Within the airfield at NASSIG (Figure 2.), many of the food sources which attract birds are water-based, mainly drainage ditches and temporary standing water during the winter. Many species of bird forage for fish, invertebrates, and vegetation that occur within these water sources. This will be covered more in the Water Sources section below.

Cattle egrets (*Bubulcus ibis*) and little egrets (*Egretta garzetta*), rock pigeons (*Columba livia*) and other bird species forage in the mixed grass and forb areas which surround large portions of the runways and taxiways at NASSIG. These areas provide seeds, vegetation, grit, reptiles, amphibians, insects, and other invertebrates that are fed upon by the various bird species depending on their needs. These areas are rarely maintained and are a mixture of short and long vegetation as well as bare ground. Maintenance consists mainly of mowing for hay production or disking vegetation, which creates fields of bare ground. Grassy areas provide food as well as cover for European rabbits, which in turn, attract predatory species such as common buzzards (*Buteo buteo*), gray herons (*Ardea cinerea*), and feral dogs to the airfield. Areas



of bare ground provide forage for seed and grit. Grit is used by many species, including rock pigeons, to aid in digestion (Figure 6).

Insect hatches occur on the airfield at NASSIG which attract large numbers of house martins (*Delichon urbicum*) and common swifts (*Apus apus*). Other bird species are also attracted to the runways to feed on insects flying in the warm air above the pavement. Rain events can trigger insect hatches or insect feeding events, which may create hazardous conditions. Insects observed includes grasshopper species, mosquitos, and other small flying insects.

Figure 6. Rock doves foraging for seed and grit in disked field.



3.6.2 Water Sources

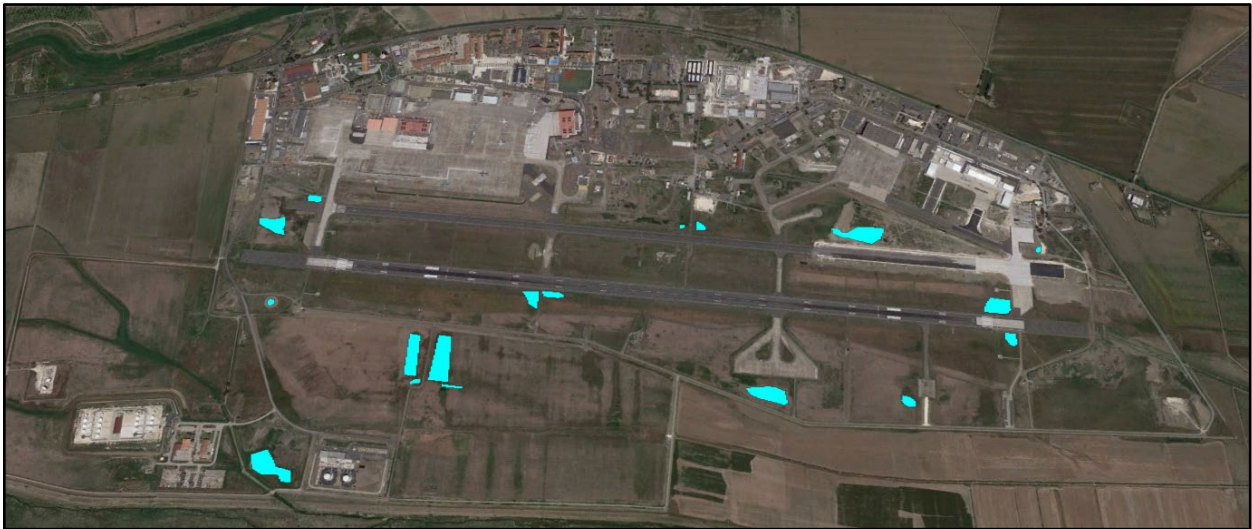
NASSIG was built between two rivers, the Dittaino and the Gornalunga, with many natural and manmade drainages flowing into one or the other (Figure 7). The drainages on the airfield are overgrown and retain water, which leads to standing water year-round creating additional habitat on the airfield that provide food and shelter. The drainages are home to turtles, fish, and invertebrates. Large (2-3 foot long) carp are regularly observed surface feeding on insects. Additionally, the topography of the airfield contains many depressions which regularly flood with rainwater in the winter, creating standing water. Since the clogged drainages cannot handle the runoff during particularly heavy rain events, over a foot of water floods the airfield, stopping aircraft operations. Flooded areas often attract wading birds, gulls, and shorebirds for loafing, drinking water, and providing them invertebrates on which to feed (Figure 8). The mapped areas of standing water is incomplete and only portrays areas seen from roads during surveys.



Figure 7. Rivers and drainages of NASSIG.



Figure 8. NASSIG Temporary Standing Water (fall/winter) (incomplete).





3.6.3 Vegetation

There are a variety of grass, forb, shrub, and tree species occurring on and near the airfield at NASSIG. Here the focus is on vegetation which provide cover to potentially problematic wildlife on and near the airfield. ITAF mows the minimum required by ICAO regulations (Figure 11). As mentioned in Food Sources above,

grassy areas provide feeding opportunities for a variety of species near the airfield at NASSIG. These grassy areas also provide cover for small bird species. Some of the more common of these include crested larks (*Galerida cristata*), wood larks (*Lullula arborea*), Italian sparrow (*Passer italiae*), rock pigeons and common (European) starlings (*Sturnus vulgaris*). In addition to the short grasses that cover most of the airfield, there are many patches of medium and long grasses interspersed. This creates “edge” habitat, which is where two habitat types meet (Figure 9). Edge habitat attracts species that use each individual habitat type along with those that are specialized to the edge habitat. A lot of the longer grasses are present due to the difficulty of mowing unlevel ground, which can be naturally occurring or it can result from construction or soil/debris dumping (Figure 10). Areas with unlevel ground would require manual mowing as the mowing tractors cannot be used in these areas.

Figure 9. Edge habitat (photos by P. Block. 2019 WHA.)

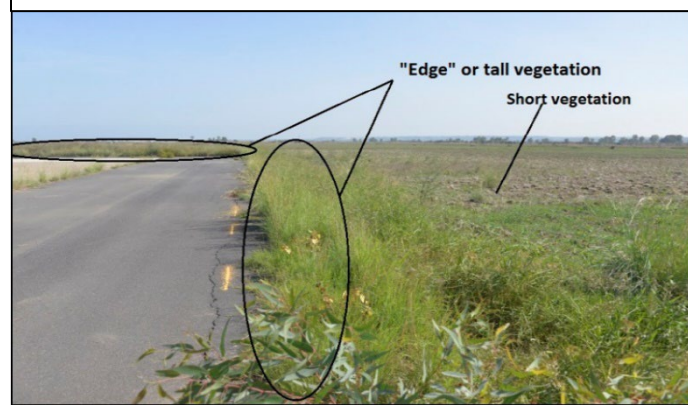


Figure 10. Debris piles (photos by P. Block. 2019 WHA.)



Shrubs and trees are also present within or near the airfield at NASSIG. They provide cover and roosting/perching opportunities for a variety of bird species. The tree east of Ramp 6 is the most problematic tree due to its usage by raptors and its proximity to the runway. Common buzzards were observed loafing in shrubs and trees on and near the airfield at NASSIG; the common buzzard is considered to be one of the more problematic species due to its potential to cause a damaging strike, due to size and occasional flocking behaviors. Common starlings roost in the trees surrounding the Tower and ITAF Ramp 5. Common

magpie (*Pica pica*) and many small passerines nest in many of the trees on the airfield and just outside the perimeter fence (Figure 12).



Figure 11. NASSIG ITAF Mowing plan (ITAF WASH Plan, 2018)



Figure 12. NASSIG Airfield trees and shrubs





3.6.4 Artificial Environment/Human Created Attractants

Elements of the artificial environment present at NASSIG, such as asphalt/concrete paved areas, buildings, and artificial structures, are potentially problematic with regard to BASH concerns because they are used by wildlife. When insects are on or just above runway surfaces, they attract large numbers of common swifts. Barn swallows (*Hirundo rustica*) and house martins have also been observed feeding on insects over the runways as well as ramp 4 and 5. The insects are attracted to the warm air created as the sun warms the pavement. Bird species, such as rock doves and yellow-legged gulls (*Larus michahellis*), use runways and taxiways for loafing, possibly for warmth. Without proper mitigation actions, these birds increase the BASH risk.

Various artificial structures near the runways at NASSIG provide perching platforms for raptors (primarily common buzzards and common kestrels [*Falco tinnunculus*]) and songbird species. While many of these structures are necessary for airfield operations, some are no longer in use. Many of the obsolete structures have been recommended for removal in previous WHA and Site Visit (SV) reports. One group of such structures is across the runway 10R/28L from taxiway Delta. These buildings are used by nesting little owls (*Athene Noctua*) and other species, including rodents. The multiple open windows and doors provide them access to the interiors of the buildings, which are covered in bird feces. (Figure 13).

Figure 13. Obsolete Airfield structures (photos by P. Block. 2019 WHA.)



Other potential BASH related issues resulting from human attractants on-base such as garbage management and feeding of wildlife have been well managed, causing little issue. However, base personnel do occasionally feed the feral dogs, likely causing them to remain on base for a longer period of time and attract others to form packs.

3.6.5 Off-site Attractants

There are several off-base wildlife attractants or potential attractants near NASSIG, which increase the BASH risk within the NASSIG airspace (strikes are considered to have occurred at NASSIG if within 5 miles of the base). These include the Gornalunga River (0.5 miles from airfield), the Dittiano River (1 mile), the Lago di Lentini (5 miles), and the Mediterranean Sea (9.5 miles). Additionally, agricultural fields and livestock pastures surround NASSIG for miles in all directions. One area of particular concern is a livestock feeding area and water feature just outside the perimeter fence to the west. Large flocks of rock doves and common starlings have been observed there and flying across the runway approach corridor to this area during airfield surveys. Offsite surveys were not conducted due to the OCONUS location, language barrier and lack of access.



Figure 14. Livestock area adjacent to NASSIG (in green).



4.0 LEGAL STATUS OF WILDLIFE

Most wildlife species are protected by European Union (EU) and Italian regulations and laws along with agreements between the USN and government of Italy. Many Italian wildlife laws codify EU Directives into Italian law. These EU Directives will be described below; their purpose is to effectively manage and maintain viable populations of specified species. Personnel responsible for managing wildlife populations at airports are required to understand and maintain compliance with the laws and regulations that relate to their duties. To ensure that personnel maintain compliance with all laws and regulations pertaining to the wildlife in question, species should be identified before any mitigation method is used. It is the responsibility of the airport to maintain compliance with all regulations.

4.1 The Birds Directive of 1979 (Directive 79/409/EEC), Amended 2009 (2009/147/EC)

The objective of the Birds Directive is to protect all naturally occurring wild bird species present in the EU and their most important habitats. It aims to control the hunting and killing of wild birds and protect their nests and eggs. It creates protected areas and improves non-protected habitats. The Birds Directive was codified into Italian law by L.157/92 on 11 February 1992.

4.2 The Convention of the Conservation of European Wildlife and Natural Habitats/ The Bern Convention (Council Decision 82/72/EEC)

The Convention of the Conservation of European Wildlife and Natural Habitats, better known as “the Bern Convention,” aims to conserve wild fauna and flora, along with their habitats, promote cooperation between nations, improve education on the need to conserve nature, and include conservation in national planning, development, and environmental policy. The Bern Convention was codified into Italian law by L.503/81.



4.3 The Convention of the Conservation of Migratory Species of Wild Animals/ Bonn Convention (Council Decision 82/461/EEC)

The Convention of the Conservation of Migratory Species of Wild Animals, better known as “the Bonn Convention,” aims to improve coordination of member states on the conservation of migratory wildlife throughout their migratory ranges.

Appendix I lists EU species threatened with extinction. Appendix II lists species that need or would significantly benefit from international cooperation. The Bonn Convention was codified into Italian law by L.42/83.

4.4 The Habitats Directive of 1992 (Directive 92/43/EEC)

The objective of the Habitats Directive is to maintain biodiversity within the EU through conservation of natural habitats of flora and fauna. It increases the protections created by the Birds Directive. EU States must create a system of strict protections for species listed in Annex II. Species in Annex II are categorized as “endangered,” “vulnerable,” “rare,” or “endemic and requiring particular attention.” The deliberate capture or killing, destruction of nests and eggs, or disturbance or destruction of breeding areas is prohibited for wildlife listed in Annex II. The Habitats Directive was codified into Italian law under Decree No. 357 of 8 September 1997.

4.5 Legislative Decree No. 281/1991

The objective of Decree No. 281/1991 is to set the standards for the treatment of companion and stray animals. Among other things, it prohibits the killing of stray dogs and cats. Exceptions are made if the animal has an incurable disease or is a proven danger, in which case only a veterinarian is permitted to euthanize.

4.6 Environmental Final Governing Standards, Italy, 7SEP2012

“The primary purpose of these Final Governing Standards (FGS) is to provide environmental compliance criteria at United States (U.S.) Department of Defense (DoD) installations in Italy. This document implements DoD Instruction 4715.5, ‘Management of Environmental Compliance at Overseas Installations,’ dated 22 April 1996, and is based upon DoD 4715.05-G, ‘Overseas Environmental Baseline Guidance Document’ (OEBGD), dated 1 May 2007.” Chapter 13 covers natural resources and endangered species. C13.3.1.2 prohibits the capture or killing; disturbance during reproduction, hibernation, or migration; and destruction or collection of eggs or nests of an Italian protected animal. The FGS implements the previously mentioned EU/Italian law in USN operations in Italy. However, the FGS does not provide language that would allow for lethal removal or trapping and relocation of species under a “Derogation” of the Habitats and Birds Directive for aviation safety, like the Migratory Bird Treaty Act (MBTA) Depredation Permit utilized at CONUS locations. The language of the FGS would need to be updated before serious conversation can occur as it would require significant cooperation between USN, ITAF, Italy and potentially the EU.



5.0 METHODS

Methods used for data collection for this WHA followed protocols and procedures from the U.S. Department of Agriculture's Wildlife Hazards Management at Airports Manual (2020), which is also based on the United States Geological Survey North American Breeding Bird Surveys protocol. All data was collected by USDA-WS Wildlife Biologist stationed at NASSIG. The data collection period for the WHA occurred from 15APR2022–15APR2023. All data was collected using paper data sheets and then entered into the Survey123 application (Environmental Systems Research Institute (ESRI) 2023).

5.1 Airfield Surveys

Airfield wildlife point-count surveys were conducted twice a month during the WHA survey period. Nineteen points were located around the airfield within the base perimeter fence. Points were surveyed in a clockwise order, proceeding around the airfield. Three minutes were spent at each station, during which all wildlife within 200 meters was recorded. Wildlife observed along the route between stations (≤ 200 meters) was also recorded as incidental observations. Rabbits were excluded from the airfield surveys. Due to their high abundance and their presence during night surveys Two survey days were completed per month, each consisting of three surveys (morning, midday, and evening). The primary purpose of the airfield surveys is to collect empirical data to monitor bird activity in the areas immediately surrounding the runways and taxiways. All wildlife was identified to species or guild (see 5.3 for definition) and counted. Behavior, habitat type, and time of observation were also recorded. See Figure 15 for a map of the airfield wildlife survey point locations. Some point locations were shifted throughout the year due to ongoing construction and the route was adjusted accordingly.

Figure 15. NASSIG airfield wildlife survey points.





5.2 Night Surveys

Monthly night wildlife surveys at NASSIG were conducted on the airfield beginning one hour after sunset. Night surveys were a point count survey using a Forward Looking Infrared (FLIR) scope. The point count survey route was followed and at each point a 360-degree scan was conducted using the FLIR. Every heat signature was identified out to 200 meters. Feral domestic animals were counted in addition to wildlife were included in observations, except for mice and rats. Rabbit observations were included in the night surveys.

5.3 Guild Classifications

To facilitate analysis, species observed during all surveys were grouped into guilds. A guild is a group of organisms that use the same ecological resource in a similar way (Merriam-Webster 2022). The guild 'mammals' did not follow the standard guild definition, as all mammals were grouped together. This was done as mammals are generally less prevalent near the NASSIG airfield. See Table 2 for a list of the wildlife guilds used and their definitions.

Table 2. Wildlife guilds used in analysis for Wildlife Hazard Assessment.

Guild	Taxonomy	Description
Aerialists	Apodiformes order or Hirundinidae family	Small, insect-eating, acrobatic flying birds such as swallow and swift species.
Blackbirds	Turdidae and Sturnidae families	The old-world world blackbirds, including blackbird species and starlings
Corvids	Corvidae family	Small to medium sized birds such as crows, ravens, jays, magpies, etc. Opportunistic feeders known for their intelligence.
Columbids	Columbidae family	Pigeon and dove species.
Grassland Birds	Motacillidae, Alaudidae, and Laniidae families	Birds normally associated with grasslands. Larks and wagtails are the commonly identified species at NASSIG.
Gulls	Laridae family	Associated with water, gulls are opportunistic, eating fish as well as scavenging many other food sources.
Mammals	Varies depending on species	Any species of mammal.
Raptors	Families Accipitridae, Falconidae, Tytonidae, Strigidae	Bird of prey species distinguished by hooked bills and talons, keen vision, and utilizing a carnivorous diet. Small to large in body size based on species. Includes: ospreys, hawks, falcons, owls, eagles, new world vultures.
Shorebirds	Charadriidae family	Small to medium bodied birds seen on coastal shorelines with long beaks for probing substrates.



Table 2. Wildlife guilds used in analysis for Wildlife Hazard Assessment (continued).

Small Songbirds	Order Passeriformes	Small to medium bodied birds. Known for various song calls, colors, and diets. Found in almost all environments.
Wading birds	Family Ardeidae	Wading birds with long legs and pointed bills commonly found in wetland and coastal habitats in shallow, slow moving waters. Includes herons, egrets, bitterns, etc.

6.0 DATA ANALYSIS

Data analyses of point count observations were conducted using Microsoft Excel® (Microsoft Corporation 2018) to determine wildlife population indices of abundance among the survey time periods and during the 12 consecutive months of this WHA. Seasonal wildlife trends were evaluated for each guild observed at NASSIG. Additionally, ArcGIS Pro (Environmental Systems Research Institute, 2021) was used to display the spatial locations of wildlife observed and to provide wildlife incidence mapping on and near the airfield. Results of these analyses are intended as spatial and temporal wildlife indices from April 2022-April 2023 and not as wildlife population estimates of abundance or density for NASSIG. For the purposes of this WHA, abundance is defined as the total number of animals counted among all observations and/or subsets (i.e., Month, Survey Period, Guild).

7.0 WHA SURVEY RESULTS AND DISCUSSION

7.1 Airfield Surveys

USDA-WS conducted sixty-nine airfield surveys during the WHA data collection period, from 15APR2022–15APR2023. Surveys were conducted on twenty-three different “survey days”, with each survey day consisting of one morning, one midday, and one evening airfield survey, for a total of sixty-nine surveys. Only one survey day was performed in July due to Regional travel requirements. The total number of individuals observed during this period was 40,204 (Figure 16-17). Wildlife of unknown species was also recorded and grouped by guild.

Airfield wildlife surveys act as an indirect hazard control measure. Assessing the wildlife present on the airfield can provide the opportunity to predict future trends and schedule aircraft operations to avoid times of high hazard. It can also provide insight on the correct times to conduct habitat management operations.

The wildlife numbers recorded during airfield surveys varied by survey period (morning, midday, evening). Among the survey periods, the morning surveys yielded the highest wildlife numbers. Midday surveys resulted in the lowest numbers, with wildlife numbers then increasing again during the evening survey period (Figure 18). Throughout the WHA data collection period, the highest wildlife numbers on the airfield occurred in November through February, likely due to migration periods and access to water during the winter. The wildlife numbers were fairly level throughout the rest of the year (Figure 19).

Species observed during the WHA data collection period on the NASSIG airfield occupied 14 guilds (Table 2). Blackbirds, consisting mainly of common starlings, were the most observed guild with 21,431



individuals observed, making up 53% of the total wildlife observed during the WHA. Columbids were the next most recorded species by guild (Figures 20-21). Approximately 99.84% of all wildlife observed during WHA airfield wildlife surveys were birds; the remainder was mainly made up of mammals, a majority of which were red foxes (*Vulpes vulpes*) and feral dogs.

Figure 16. Observed wildlife recorded on NASSIG during WHA data collection.

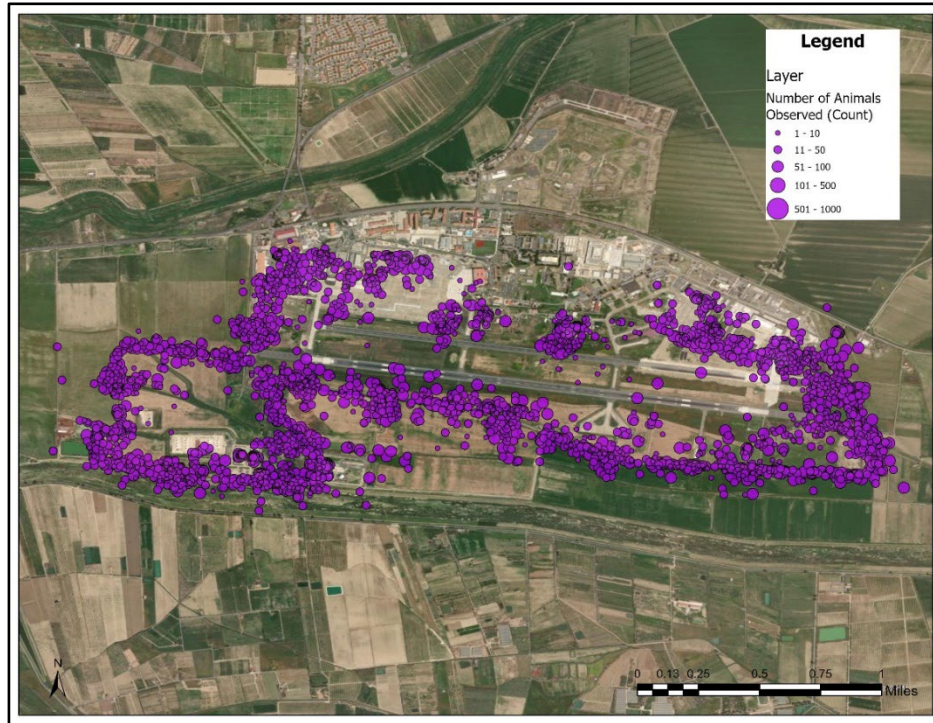


Figure 17. Averages of wildlife recorded on NASSIG during WHA data collection.

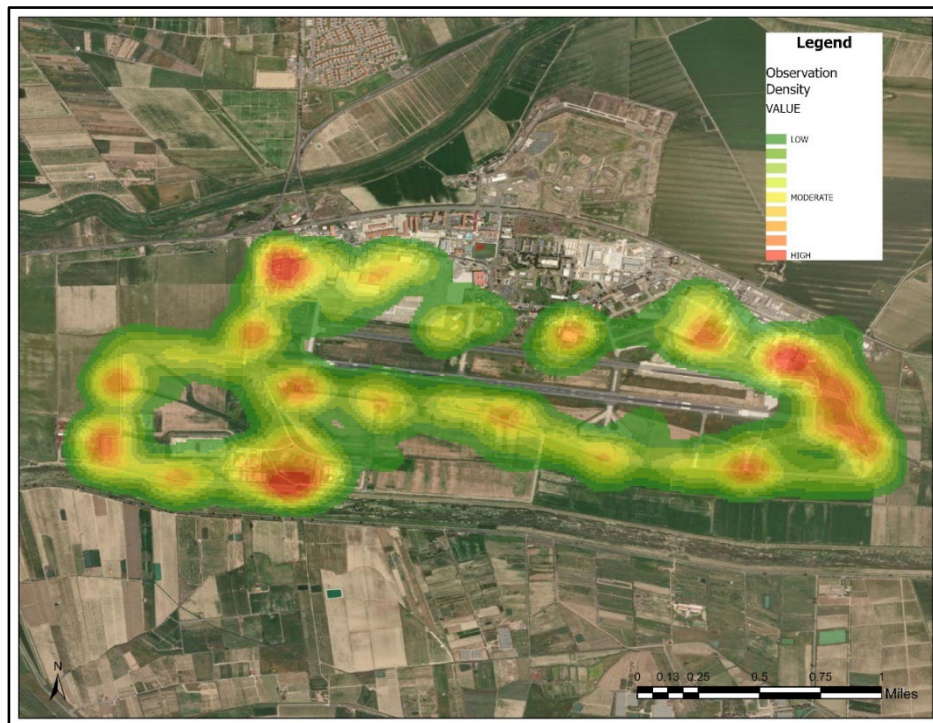




Figure 18. NASSIG WHA airfield surveys – Total wildlife recorded by survey period.

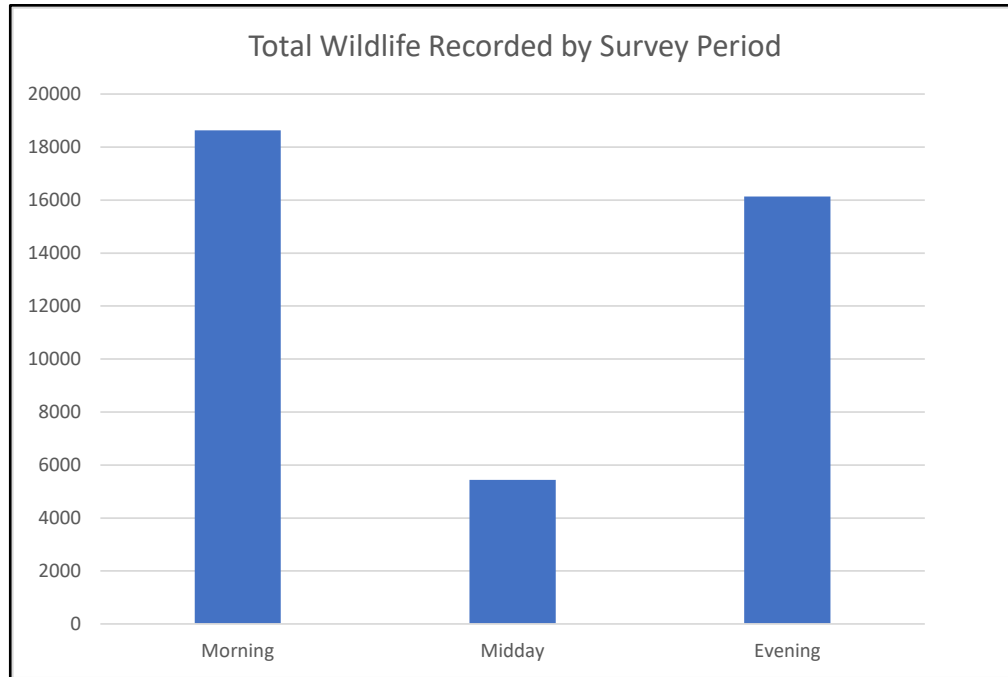


Figure 19. NASSIG WHA airfield surveys – Total wildlife observed per month.

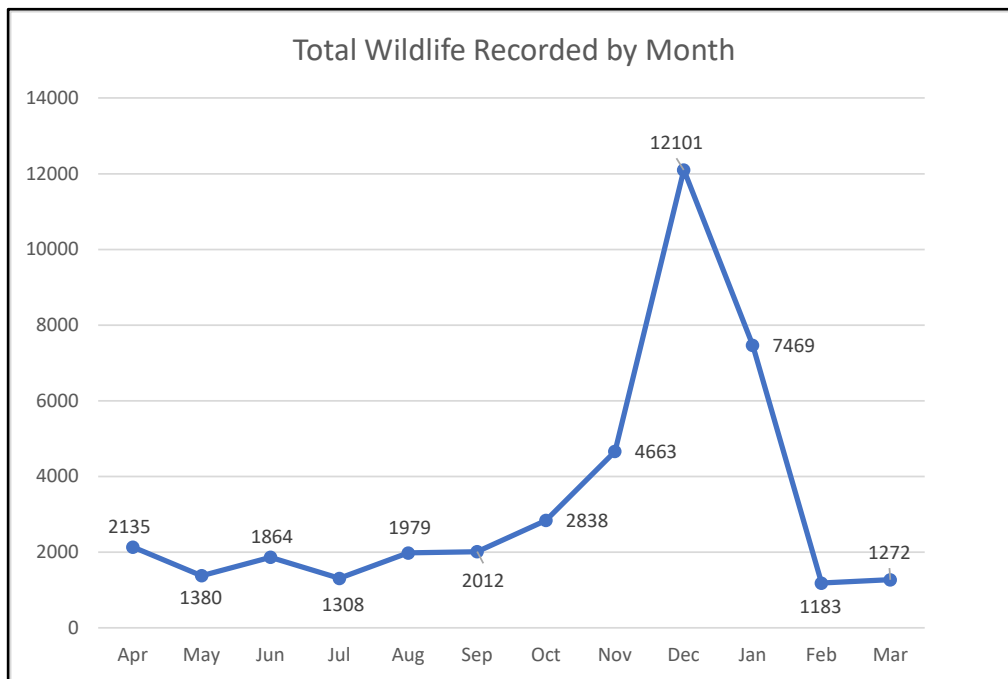




Figure 20. NASSIG WHA airfield surveys – Total wildlife observed per guild.

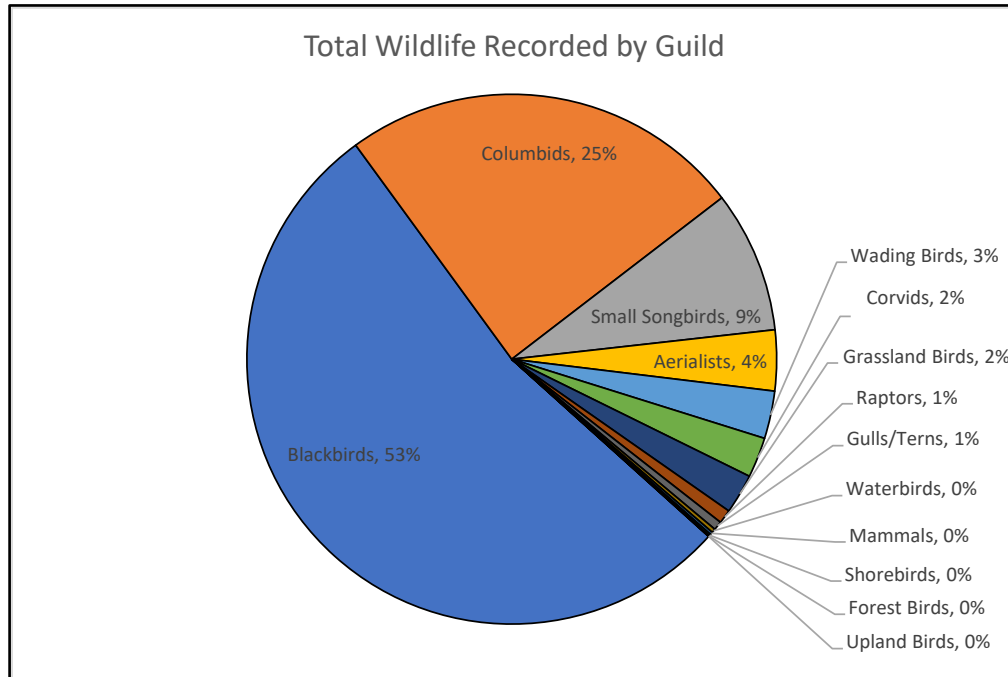
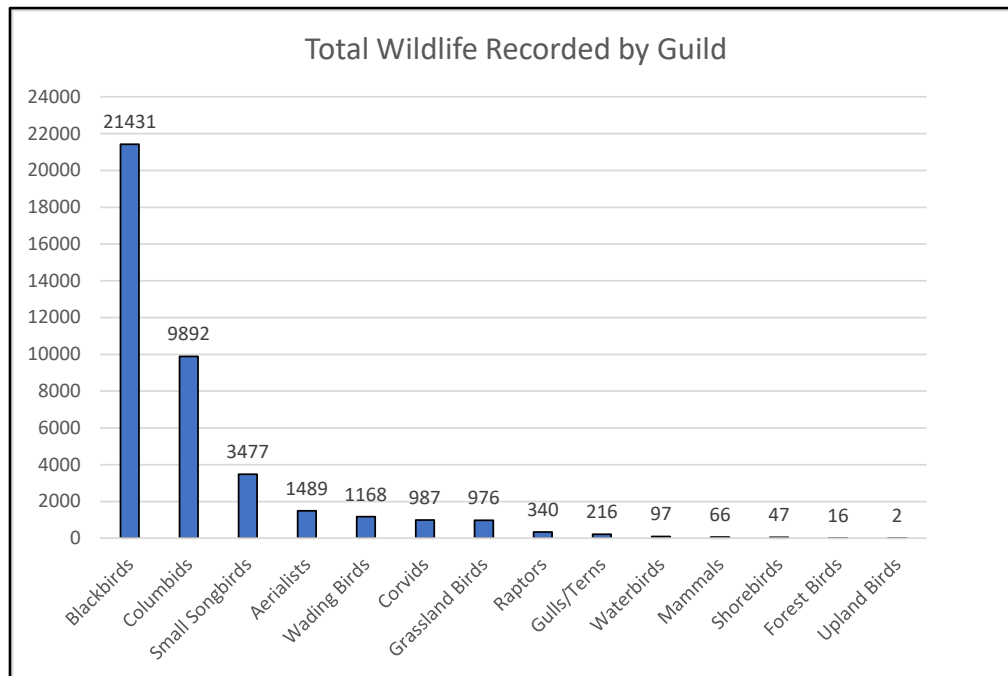


Figure 21. NASSIG WHA airfield surveys – Wildlife totals per guild.





Of the fourteen wildlife guilds observed during data collection, seven guilds were identified as having great seasonal variation in terms of population numbers and/or as having species occurring within them that are especially hazardous on the NASSIG airfield. These seven guilds all consisted of birds. While most of the sixty-one species of wildlife observed during the WHA data collection period (Appendix 3) may be hazardous to aircraft, species among these seven guilds represent a greater hazard at NASSIG due to large numbers, frequent airfield crossings, body size, or a combination thereof. These seven guilds include blackbirds, Columbids, small songbirds, aerialists, wading birds, raptors, and gulls. Figure 22 attempts to show how the populations of these species varied seasonally on the NASSIG airfield during the WHA data collection period. The species observed during the WHA which represent a comparatively smaller threat to aircraft, occupied the following guilds: Corvids, grassland birds, waterbirds, mammals, shorebirds, forest birds, and upland birds. While species within these guilds may cause damaging wildlife strikes, observations on the NASSIG airfield indicate that the airfield presence of species within these guilds have a significantly lower associated risk in comparison to that of the species within the seven guilds that are represented in Figure 22. Figure 23 displays the relative numbers and seasonality of all guilds observed during the WHA. Wildlife guilds will be discussed in greater detail in **7.4 Guild Observations** (below).

Figure 22. Wildlife recorded at NASSIG during WHA surveys – Problematic guilds.

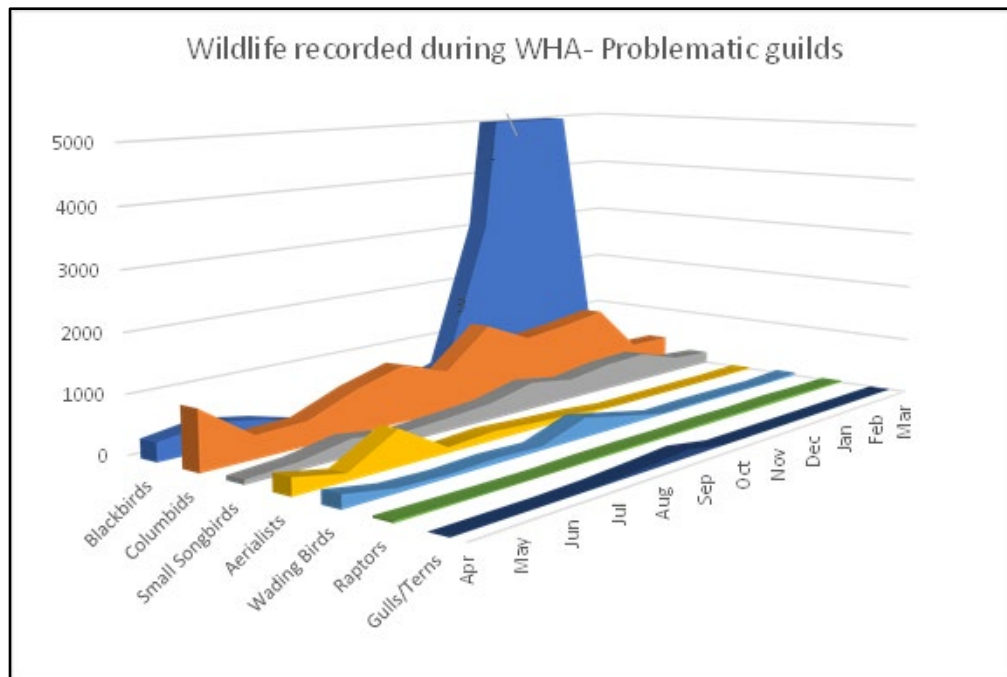
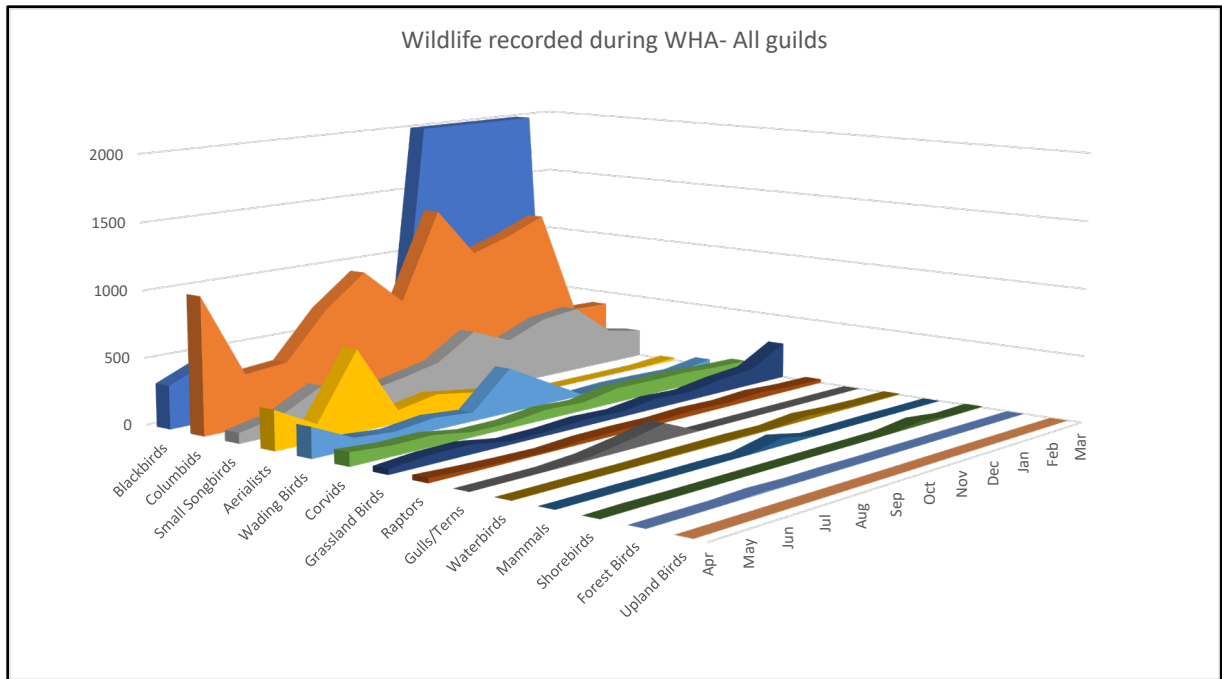




Figure 23. Wildlife recorded at NASSIG during WHA surveys – All guilds (# of Individuals observed).



7.2 Night Surveys

Night survey results during the WHA period shared similarities with airfield survey results. Night wildlife observations were high from November–April, with the highest number of observations recorded in December (Figure 24-25). Mammals, specifically European rabbits, were the most observed guild during night surveys (Figure 25). European rabbit populations are present in very high densities and are only controlled by disease. Bats were not recorded during night surveys during the WHA data collection period but were observed. Barn owls (*Tyto alba*) were only observed during night surveys at NASSIG.

Figure 24. NASSIG WHA night surveys excluding European rabbits.

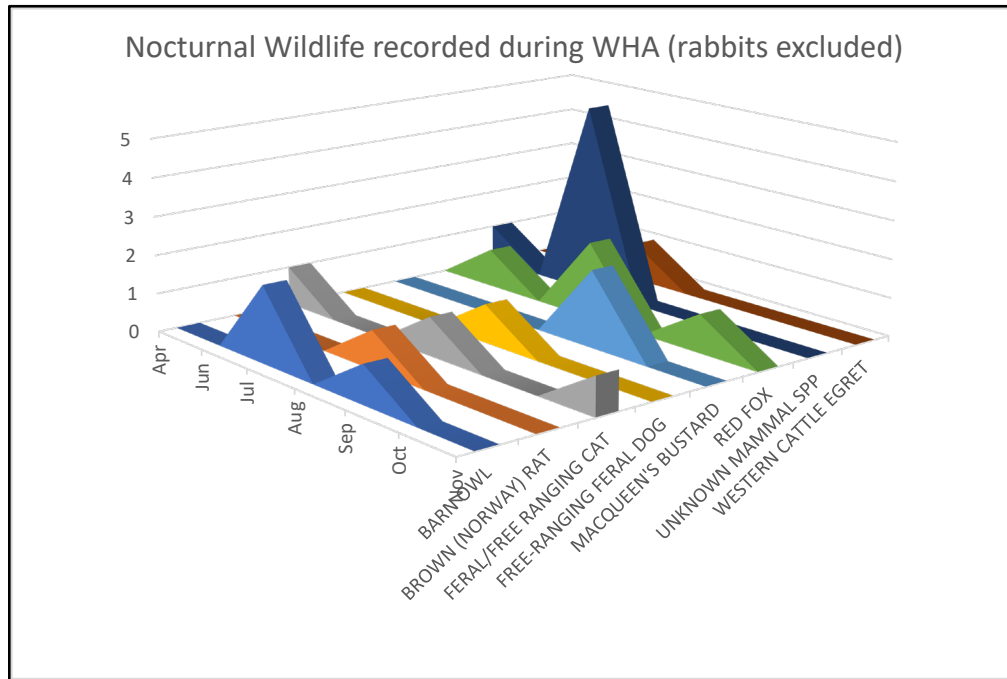
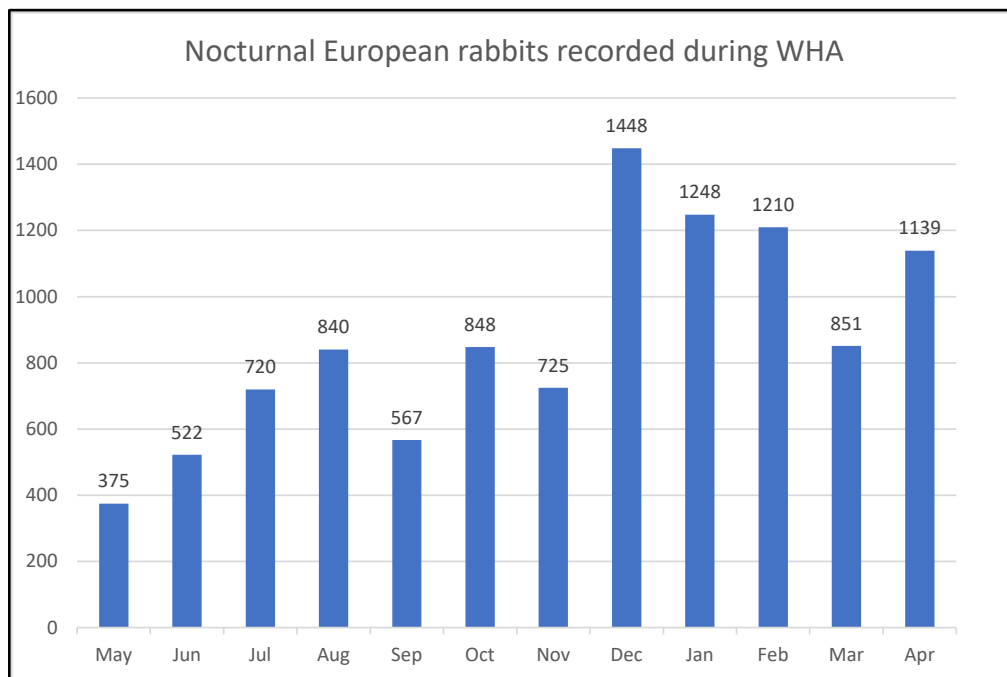


Figure 25. NASSIG WHA night surveys (European rabbits).





7.3 Guild Observations

7.3.1 Aerialists

Description: This guild is represented by martins, swallows, and swifts. Examples of these birds that were observed at NASSIG during the WHA include house martins, barn swallows, and common swifts. Aerialists travel in flocks and can fly erratically across airfields hawking (aerial foraging) in search of food, which mainly consists of insects. This behavior makes them a commonly struck bird at certain times of the year, mainly April through June.

Attractants: Aerialists feed upon flying insects and are often seen flying in flocks over grass fields and large concrete/asphalt areas. Grasses that have been allowed to grow taller than is recommended at airports are of particular concern as the number of insects is greater in such habitat, serving as a greater attractant to aerialists. Aerialists are also commonly observed over the drainages along Ramp 4 and off the western end of the runway feeding on the associated insects. It is also important to note that on the cooler/overcast days and mornings at NASSIG, aerialists can be seen heavily foraging over the concrete and asphalt areas of the airfield, specifically Ramp 4 and the 28L Approach underrun. These birds follow the large insect clouds that tend to linger over these areas for the radiant heat. Once the temperature rises, the insects move to grassy areas and the birds follow.



Common swift



Western house martin

Although evidence of aerialists nesting on NASSIG has not been observed, large structures such as hangars, maintenance buildings, and bridges provide great nesting options for aerialists. Many of these species create nests out of mud, usually in the corners of high structures.

Risks: Nationally, this guild represents 11.4% of known birds struck by civil aircraft from 1990–2022 in the U.S. (Dolbeer et al. 2023). During the WHA at NASSIG, aerialists made up a large portion of the non-damaging aircraft strikes and they were involved in 17 of the 46 (36.9%) total reported aircraft strike events. Aerialists were also involved in 11 other non-damaging strikes in the ten-year strike record.

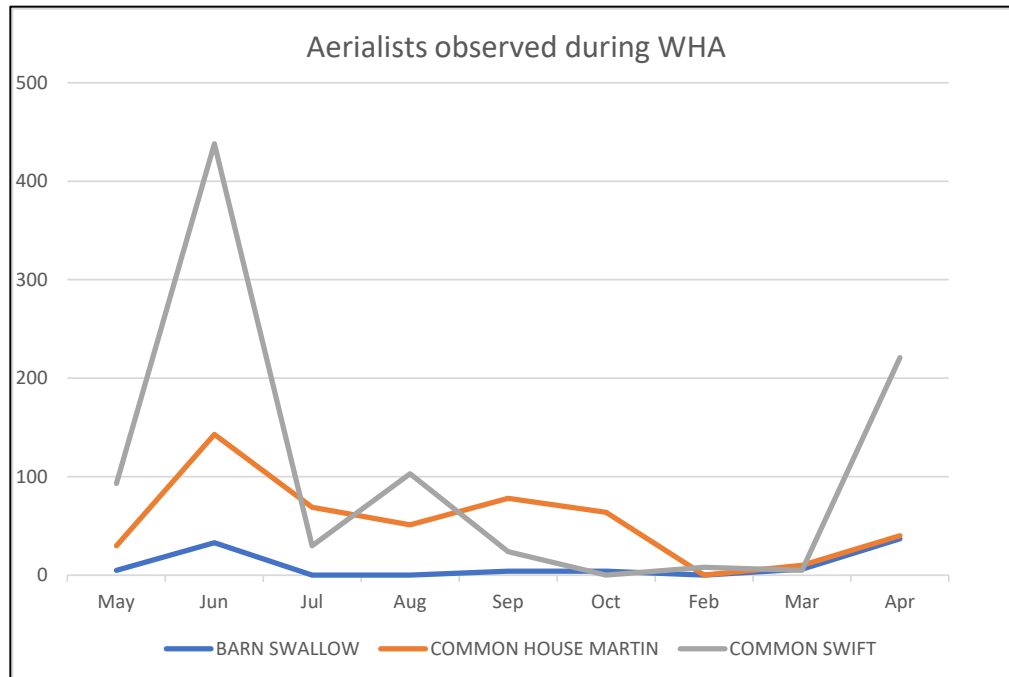
Legal status: Aerialists are protected by L157/92 and are listed in Appendix II of the Bern Convention. As such, it is also protected by the FGS Italy.

Control measures: Swallows and swifts may be best managed through an integrated program that includes habitat management, insect control, and harassment. If insect populations spike, insecticides are a possible management option. When aerialists are present, hazing them off the field using pyrotechnics has limited short-term success. Birds will often simply move to another location on the airfield, so it is important to be persistent in hazing. Aerialists have not been observed nesting on NASSIG but any future nesting areas should be identified and exclusionary devices be installed to prevent future nesting

Prevalence: Aerialist numbers on the airfield peaked in June with high numbers being recorded from Apr-Jul (Figure 26). Most aerialists observed seemed to be drawn to the airfield to feed on periodic insect hatches. This species group was present in large numbers, elevating BASH, particularly during the morning hours and cloudy/overcast days at NASSIG.



Figure 26. Aerialists recorded during WHA airfield surveys at NASSIG.



7.3.2 Blackbirds

Description: The term blackbird loosely refers to two different species of starling and a species of blackbird. The species seen on NASSIG during the WHA were the common starling and the spotless starling (*Sturnus unicolor*). These species were also observed outside the WHA period in addition to the common blackbird (*Turdus merula*).

Blackbirds are a varied group that possess conical, sharp-pointed bills and have rather flat profiles. All blackbirds are gregarious, especially in winter when thousands may roost together, often in a flock of mixed species. In NASSIG's geographic area, large flocks of blackbirds begin to form roosts as early as October and disband by February (Figure 27). Blackbirds are active during daylight hours.



Common starling, Photo by W. de Vries

Attractants: Blackbirds are omnivorous, feeding on grains, weed seeds, fruits, and insects. Blackbirds are generally found in urban areas, airports, grassy or weedy fields, and fallow croplands. Abandoned buildings and trees planted on or adjacent to airports serve as potential roosting sites. Starlings prefer croplands and weedy fields; they are known to roost and nest in marshy areas with tall vegetation.

Risks: Blackbirds are considered a serious threat to aviation because of the large flocks they form and the density of the bird. Nationally, blackbirds account for 6.1% of all civil aircraft bird strikes of known species (Dolbeer et al. 2023). Common starlings are rated as the sixth most hazardous species (Devault et al. 2018). Additionally, their winter roosts present a nuisance because of the noise and the associated droppings that corrode and damage buildings and property. If allowed to accumulate, droppings can become a source of diseases that can infect humans and domestic animals. Blackbirds, specifically common starlings, were involved in one non-damaging strike event at NASSIG during the WHA and are not found in the ten-year strike record. During fall/winter migration, very large flocks (up to approximately



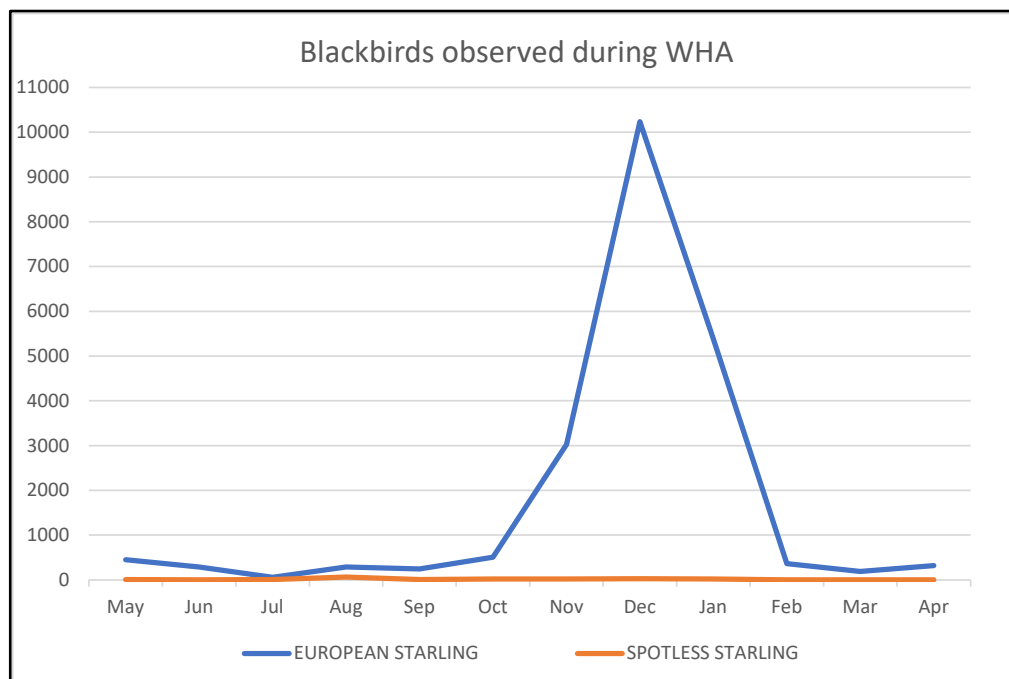
5,000+ individuals) were seen flying over the airfield, usually around the runway 28R approach. The longer the birds remain on or near the Airfield Operations Area (AOA), the greater the chance that they will interfere with air traffic. There was one European starling strike identified during the WHA period. There were no other blackbirds among the known species to be struck in the ten-year strike record.

Legal status: Blackbirds are protected by L157/92 and as such they are also protected by the FGS Italy. Spotless starlings are also listed in Appendix II of the Bern Convention.

Control measures: Habitat management (i.e., grass management, roost removal, etc.) is usually the most cost-effective management because it serves as a long-term deterrent. However, grass management has mixed results in controlling blackbirds utilizing an airport. Regular mowing is wise management for all accessible areas of the airport (Cleary et al. 2005). The management of roosting sites on or adjacent to an airport will also reduce blackbird presence. Habitat management alone may not be enough to discourage bird presence. When blackbirds are present, airport personnel should haze them off the field using pyrotechnics. Flocks of birds can be dispersed with pyrotechnics and visual repellants. Birds will often simply move to another location on the airfield, so it is important to be persistent in hazing any bird species.

Prevalence: Due to their large flock size, blackbirds were the most frequently observed avian guild, comprising 53.1% of the observations on the airfield during the study period (Figure 20). Blackbird species were most prevalent at NASSIG during the months of October through February but were also observed in smaller numbers during the remaining seven months of the survey. The fall/winter peak in blackbirds was likely due to fall migration as well as breeding and raising young. Throughout the year, this guild was observed primarily in short grass habitat and in the pastures off base.

Figure 27. Blackbirds recorded during WHA airfield surveys at NASSIG.





7.3.3 Corvids

Description: The only Corvid species observed at NASSIG during the WHA was the common magpie. Hooded crows (*Corvus cornix*) were observed during previous WHAs. These birds are medium to large in size with long legs, thick necks, and heavy, straight bills. In flight, the wings are broad and rounded with the wingtip feathers spread like fingers.



Common magpie, Photo by J.C. Salvadores

Attractants: Corvids usually prefer open grasslands and marsh areas. They are omnivores, foraging on many different foods ranging from fruits and vegetables to eggs and carrion. The agricultural fields surrounding NASSIG are a great attractant for corvids. There is a large number of nesting bird species around the airfield and corvids are often attracted to their nests/eggs. The numerous European rabbit carcasses around the airfield are also frequently scavenged by magpies. Common magpies are the only bird observed nesting on NASSIG during this WHA; their nests were observed in the razor wire of the perimeter fence and in structures around the airfield. Other species are likely to nest on NASSIG but were not identified.

Risks: Corvids are generally regarded as the most “intelligent” bird guild and are rarely struck. However, due to their medium to large size, these birds do pose a risk to aircraft. Corvids are gregarious in nature; larger flocks of these birds can pose a greater risk when near airfields. Corvids were not struck during the WHA but were involved in one non-damaging strike in the ten-year strike record.

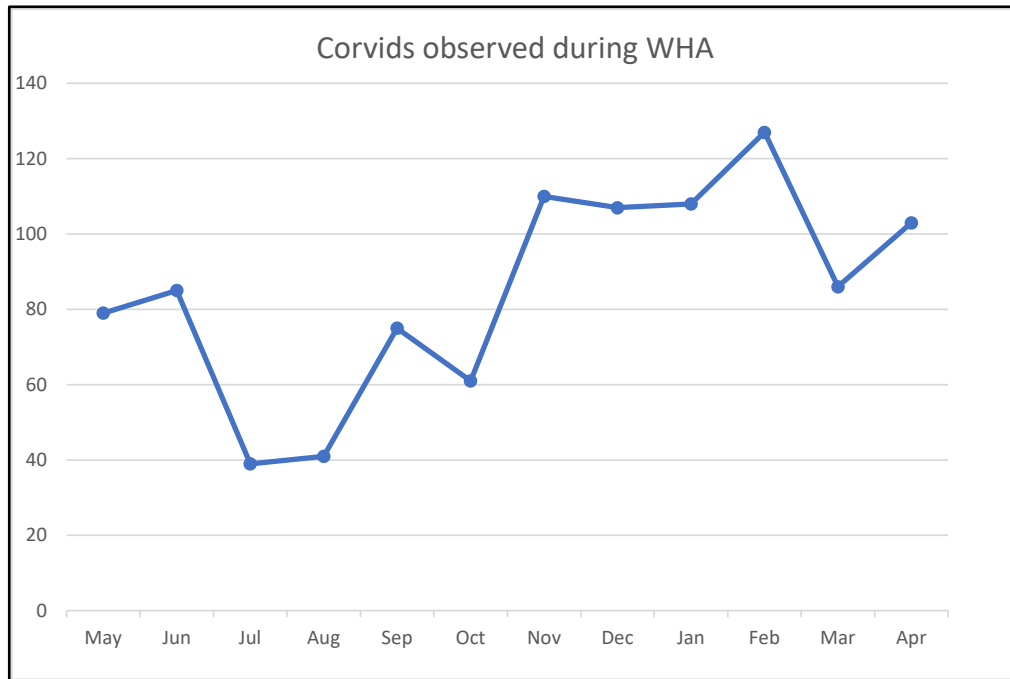
Legal status: Magpies are listed in Appendix 2/I of the Birds directive. There is a hunting season for licensed hunters in Italy from September through January.

Control measures: Corvids may be best managed through an integrated program that includes habitat management, carrion removal, and harassment. Habitat should be managed so that the grass has a uniform composition, density, and height (Cleary et al. 2005). Agricultural activities near the airfield should be avoided, or frequently monitored if it cannot be. Any unused taller structures at the edge of the AOA, which corvids are observed using, should be removed. Spike strips can be installed on structures around the airfield to prevent corvid perching. If specific corvids habitually frequent an area, they should be routinely dispersed with pyrotechnics, vehicles (e.g., horns) and human presence. Due to their intelligence, a single form of harassment is usually ineffective long-term. Since magpies have been observed reusing nests on the airfield, unused nest should be removed.

Prevalence: Corvids were relatively non-existent in major densities around the airfield, but were observed year-round (Figure 28). However, due to their potential to cause damaging strikes, continual monitoring is recommended.



Figure 28. Corvid recorded during WHA airfield surveys at NASSIG.



7.3.4 Columbids

Description: Pigeons and doves are powerful fliers with robust bodies, small heads, and short beaks. Eurasian collared doves (*Streptopelia decaocto*) and European turtle doves (*Streptopelia turtur*) typically fly close to the ground near cover as they travel between feeding and roosting areas. Wood pigeons (*Columba palumbus*) and rock pigeons tend to fly at higher altitudes, descending to their destinations in a rapid circling pattern with wings spread back. Although columbid species are primarily granivorous, they will occasionally consume protein-rich animal material, such as insect larvae. Pigeons are also opportunists, feeding upon handouts from humans.



Attractants: Agricultural and short grass habitats provide feeding and loafing areas for columbids. The disking of grassy fields around the airfield brings seed to the surface and provides grit. Grit is dirt, sand, and small rocks that are ingested and held in the crop to help break down seeds and other food sources, aiding in digestion. When these disked fields are then seeded, as was observed in December of 2023, it provides even more forage for all bird species. The presence of grass seed and disked fields close to the AOA could increase the potential hazard presented by the guild. The livestock feedlot to the southwest of the airfield, in line with the runway 10R approach, attracts large flocks of rock and wood pigeons, in addition to the aforementioned starlings. Monitoring of this area should be a priority. Care should be taken during construction activities to avoid extended exposure of bare ground. The collared doves and turtle doves recorded at NASSIG were primarily observed in the south-western corner near the





Defense Reutilization Marketing Office (DRMO) lot. Shrubland habitat in this area provides nesting and loafing habitats. Buildings can provide desirable nesting areas (e.g., flat surfaces and ledges, metal I-beams in hangars, etc.) for pigeons.

Risks: Columbids are considered a serious threat to aviation because of the large flocks they form and their large bodies. Nationally, columbids account for 13.2% of all civil aircraft bird strikes of known species (Dolbeer et al. 2023). Rock pigeons are rated as the fourth most hazardous species (Devault et al. 2018). Columbids also present a nuisance because of the associated droppings that corrode and damage buildings and property. If allowed to accumulate, droppings can become a source of diseases that can infect humans and domestic animals. A rock pigeon, was involved in one strike event at NASSIG during the WHA with columbids being recorded as involved in nine other strike events in the ten-year strike record that includes one Class D damaging strike with damaged estimated at \$39,435.

Legal Status: Rock pigeons and collared doves are protected by L157/92 and are listed in Appendix 2/I of the Birds directive and Appendix III of the Bern Convention. Wood pigeons are listed in the Birds directive Appendix 2/I and 3/I. There is also a legal hunting season from September through January. Turtle doves are listed in the Birds directive Appendix 2/II and Appendix III of the Bern Convention. There is also a legal hunting season from September through December.

Control Measures: Habitat modification would help reduce the number of Columbids directly using the airfield. Weedy fields should be cut prior to seed development and/or replaced with grass.

Disking of the airfield, in its current iteration, is generally not an acceptable solution for vegetation management as it causes more problems than it solves. The disking (and seeding) is likely occurring for the commercial production of hay on base by a local farmer.

Newly constructed structures should be designed to exclude nesting. Old buildings should be retrofitted, where feasible, with exclusionary netting or other type of barrier to block access to eaves and beams. It is important to be diligent with control efforts because these birds will return and attract additional birds to the site.

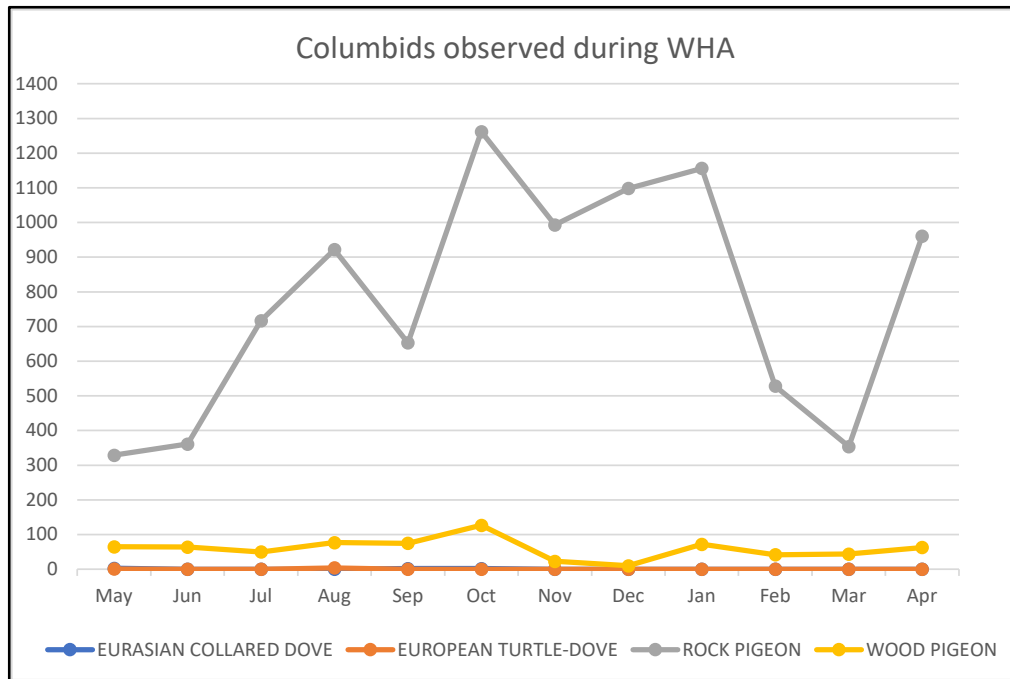
There is a legal hunting season for rock pigeons and turtle doves that generally runs from September through January (December for turtle doves). While legal hunting for some species can be an effective tool to reduce bird strike hazards at airports, there are often accessibility restrictions, security concerns, harvest limits, and the hunting season may not coincide with the time that control is needed. Hunting would also need to be accomplished by local nations and not by USDA-WS or USN personnel, due to restrictions in the FGS.

Pyrotechnics can be an effective deterrent for these species, particularly rock doves, but tend to be a short-term solution. If desirable habitat elements exist, Columbids generally return.

Prevalence: Columbids were regularly seen at NASSIG throughout most of the year (Figure 29). As previously stated, columbids represented the second most commonly observed guild during the WHA survey period, with rock pigeons being the most observed of the guild. Columbid sightings were typically in grassy areas. They were also seen on/around fences, power lines, and utility structures. These are favored perching locations for columbids and it makes them easily visible during surveys. Rock pigeons were observed entering and loafing in the hangars during nearly all WHA surveys. Large flocks of rock doves were regularly observed flying over the airfield during the morning surveys.



Figure 29. Columbids recorded during WHA airfield surveys at NASSIG.



7.3.5 Grassland birds

Description: The grassland birds observed during the WHA survey period at NASSIG consist of five species: crested lark, gray wagtail (*Motacilla cinerea*), white wagtail (*Motacilla alba*), wood lark, and woodchat shrike (*Lanius senator*). This guild consists of medium-small sized birds that utilize grassland as their primary habitat. Grassland birds forage for seeds and insects. They are classified separately from “small songbirds,” due to their size and behavior.



Crested lark

Larks and wagtails tend to be resident birds in Sicily and are not prone to migration. Changes in the numbers observed are likely due to seasonal habitat changes and breeding/fledging (when birds leave the nest) of young.

Attractants: Agricultural and short grass habitats provide feeding and loafing areas for grassland birds. The seeds and insects in the grasses are the primary attractant for grassland birds. Wagtails were often observed foraging on the perimeter road. Crested and wood larks utilize the perimeter fence, structures, and signposts to perch and vocalize. These vocalizations are used to claim territories and attract mates. There was one woodchat shrike observed that was likely migrating through and is not considered a common species.



White wagtail, Photo by I. Davies

Risks: Grassland birds are small-bodied and rarely form flocks. Although they are not as large as many other species considered detrimental to air safety (e.g., wading birds, gulls, raptors), they are still a concern because of their overall abundance, dense body structure, and movements around the airfield. One red-backed shrike (*Lanius collurio*) was reported to have been struck by an aircraft at NASSIG during the WHA period and one non-damaging strike of a crested lark is recorded in the ten-year strike record.

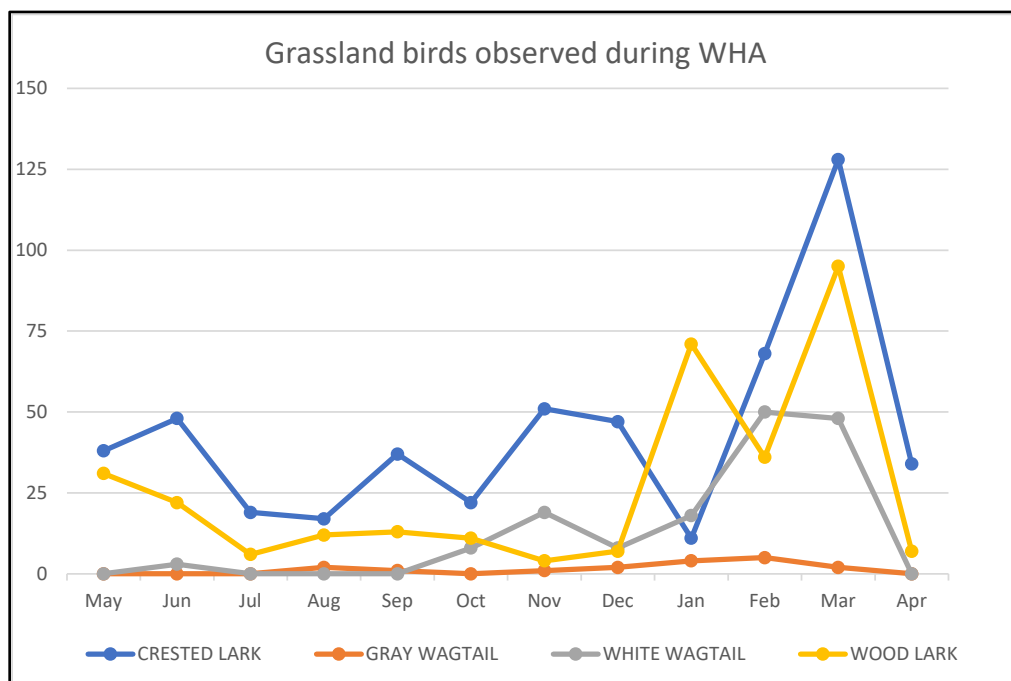


Legal Status: Grassland birds in general are protected by L157/92 and other different laws and acts. Woodchat shrikes, gray wagtails, and white wagtails are listed in Appendix II of the Bern Convention. Both crested larks and wood larks are listed in Appendix III of the Bern Convention, while wood larks are also listed in Appendix I of the Birds Directive.

Control Measures: Habitat modification helps to reduce the number of grassland birds directly using the airfield. Weedy fields should be cut and/or replaced with grass. All vegetation should be mowed before being allowed to seed. Habitat management (i.e., grass management, roost removal, etc.) is usually the most cost-effective management because it serves as a long-term deterrent.

Prevalence: Grassland birds were observed year-round on the airfield (Figure 30), with a spike between January and April. This spike likely correlates to the breeding season of crested and wood larks when they establish territories by vocalizing and displaying from elevated perches. This is followed by the hatching and fledging of young. Wagtails, both white and gray, were observed in stable numbers throughout the year. A single woodchat shrike was observed in July.

Figure 30. Grassland birds recorded during WHA airfield surveys at NASSIG.





7.3.6 Gulls

Description: Gulls are robust birds with webbed feet, long, pointed wings, and a stout, slightly hooked bill. Most adult gulls are white with gray backs and black wing tips. Juvenile gulls are typically a mottled brown color with black bills.

The only gull species positively identified on the airfield at NASSIG during the WHA data collection period was the yellow-legged gull (*Larus michahellis*). Many gulls were observed, but could not be positively identified to species because they flew over the airfield at high altitudes.



Yellow legged gull, Photo by J. Marques.

Attractants: Gulls are attracted to water and food on and around airfields. Gulls are omnivorous with a diet including refuse from dumpsters and landfills, earthworms, insects, fruit, grain, and carrion. The presence of a few gulls can act as a strong attractant to others passing by. Gulls can be extremely gregarious and if they observe one feeding, that one will decoy others in to the area. Agricultural tillage on and around the airport can be an attractant to gulls, who feed on the exposed worms and potentially on crops as well. Gulls can be readily observed flocking on runways or taxiways following rainstorms to feed on earthworms.

They are attracted to the ideal loafing sites that airports often provide. Any ponding found on the airport will attract gulls as a secure place to drink and rest. Gulls will remain in ponding or sitting in a comfortable area with food for as long as they remain left alone.

Risks: Gulls have the distinction as the bird most reported in damaging strike incidents in the U.S. They are generally considered a significant hazard due to their size, abundance, wide and expanding distribution, flocking behavior, relatively slow flight characteristics, and general tendency to concentrate at airports.

During the WHA data collection period, one Mediterranean gull (*Ichthyaeetus melanocephalus*) and one black-headed gull (*Chroicocephalus ridibundus*) were each involved in a separate non-damaging strike event, but both were likely struck while over the Mediterranean Sea.

Legal status: Gulls are generally protected by L157/92. Yellow-legged gulls are protected by L157/92 and are listed under Appendix III of the Bern convention.

Control measures: Habitat modifications, such as the removal of refuse and carrion (i.e., animal carcasses) from the airfield and the elimination of standing water (increased drainage), are among the most effective methods of gull control. Gulls can habituate rather quickly to hazing (pyrotechnics, bioacoustics, and visual scare devices). Gulls should be hazed early and often in any area of the field in which it is safe to do so. A zero-tolerance policy prevents the birds from becoming accustomed to using the airfield.

Prevalence: Unidentified gulls were recorded as unknowns within the guild. Of gulls observed, this category, gull (unknown), had the highest numbers recorded. This was due to the difficulty in distinguishing between gull species. Gulls were only observed between June and September during the WHA and were always observed flying over the field.



7.3.7 Mammals

NASSIG hosts a variety of mammal species. For the purposes of this section, mammals were divided into two subsections: feral domestics (feral dogs) and small mammals. Feral dogs are the largest terrestrial mammal at NASSIG and due to their activity in the immediate vicinity of the airfield, they represent the greatest potential of damaging aircraft strikes with mammals. Small mammals, though abundant around NASSIG, create a more indirect BASH hazard via the attraction of larger predators, such as feral dogs and raptors.

Feral Dogs

Description: Feral dogs are common on the island of Sicily, especially around populated areas. Some dogs observed on the airfield may be free-ranging dogs owned by local farmers. They come in a wide variety of shapes and sizes and can weigh between 20–70 lbs. The most frequently observed are medium to large mixed breeds. Feral dogs are mainly nocturnal, but may be observed any time of day. Feral dogs are scavengers and will eat almost any animal or vegetable matter. They hunt for rodents and rabbits and will cache uneaten food.

Attractants: Feral dogs are commonly found on NASSIG foraging for rodents and rabbits that thrive in open grassland habitats. Certain feral dogs are also fed and cared for by local nationals.

Risks: Due to their large body size and tendency to travel and hunt in packs, feral dogs pose a severe threat to aircraft and they have also been documented to attack personnel on the airfield.

Legal status: Feral dogs are protected by Legislative Decree No. 281/1991. Feral dogs cannot be relocated by USN and can only be handled by approved HN personnel. There was a prior agreement between NASSIG and the commune of Lentini to remove a specified number of feral dogs from the airfield every year. However, the commune of Lentini has ended this agreement, leaving USN with no legal options at the moment.

Control measures: Exclusion is the most effective method to reduce risk. Feral dogs can be excluded from an area with a wildlife deterrent perimeter fence. The fence must be tall enough that the dog cannot jump it and must either be buried or modified by other means at the base so the dogs cannot dig under (Appendix 6). Gate security, provided by USN or ITAF, would also be required to harass any nearby dogs away from the main gate.

Prevalence: Feral dogs were seen within the AOA on a limited basis (2 occasions) during WHA survey periods. However, there was evidence of their presence (e.g., tracks and scat) as well as numerous other sightings outside of survey periods.

Wild Mammals

Description: European rabbits and red fox are common at NASSIG, especially at dawn, dusk, and overnight. European rabbits are in the lagomorph family and weigh approximately 1.5 kg, are mostly grayish brown, lighter under belly, with ears tipped dark brown. Other small mammal species present on or near the NASSIG airfield include rodent species such as mice and rats. Stoats (*Mustela erminea*) have also been observed during previous WHAs but were not observed during this WHA time period.



European rabbit, Photo by W. Siegmund



Red foxes are a medium size canid that weights approximately 5.5 kg to 6.5 kg. Its color varies but is usually a rusty red with a reddish black tail and white chest. Red fox tend to be shy and prefer to avoid human interaction.

Bats are small flying mammals that are present on the airfield during the night hours. Bats feed on flying insects.



Red fox

Attractants: Small mammals utilize grassland areas, shrubs, and ornamental plants for feeding. During the day, they typically stay close to areas that provide visual obstruction from predators (i.e., buildings, shrubs, and tall grasses). Rodents, especially mice and rats, will eat insects as well and will chew through different materials to access food, water, and create nests.

Red foxes are omnivores, feeding primarily on rabbits, rodents, birds and other sources or meat including carrion. Red fox will also feed on eggs, berries and trash. Edge habitat is preferred hunting zone for red fox.

Bats are attracted to flying insects. These insects are usually attracted to lights which are ever present on both the airfield and aircraft.

Risks: Wild mammals create very little direct risk to aircraft due to their small body size. However, small mammals can cause damage to landing gear if they are struck during takeoff/landing. They also pose an indirect risk of damage to stationary planes (foraging for wires, insulation, and other components for food/shelter), hangars, and other buildings/structures. They present a significant indirect risk to aircraft as they are a major prey item for raptors (13% of all strikes in the U.S.). One European rabbit was struck by an RQ-4 in 2012, causing damage to the landing gear in a Class C strike event

Rabbits and rodents may also chew wires, effecting lighting and signage on the airfield. They also provide an attractant to red fox as an optimal food source.

Bats pose a risk similar to that of birds of a comparable size. There were four non-damaging bat strike events, all Savi's pipistrelle (*Hypsugo savii*), at NASSIG during the WHA period, and an additional Savi's pipistrelle and Kuhl's pipistrelle (*Pipistrellus kuhlii*) were recorded in the ten-year strike record in non-damaging events.

Legal Status: Red foxes and European rabbits are generally protected by L157/92. Stoats, Savi's pipistrelle and Kuhl's pipistrelle are protected by L157/92 and listed in Appendix III of the Bern Convention.

Control Measures: If habitat is not maintained, small mammal populations can rapidly increase, attracting other unwanted wildlife to the airport. Habitat management, such as mowing grass and reducing brushy areas on airfields, will reduce shelter and food for small mammals. By reducing the presence of small mammals on the airfield and reducing edge habitat will reduce fox presence on the airfield

In terms of bat management around the airfield, harassment is not a viable option as they are nocturnal and are not easily swayed by pyrotechnics and/or human presence in general. Insect management is important, especially with the presence of standing water in drainages. Insects can become quite numerous during certain times of the year at NASSIG; insecticide treatments, applied at least twice a year, would help manage the insect population and the bats that feed upon them.



Prevalence: Small mammal observations were recorded during monthly nocturnal surveys (See section 7.2 above). High densities of European rabbits were observed every month of the WHA, with the highest counts being between December and April. Without some form of control, rabbit populations will continue to remain dense, providing a food source for local raptors, scavengers, and feral dogs.

7.3.8 Raptors

Description: Raptors are predatory, scavenging birds that feed upon prey animal species. Raptors include vultures, eagles, hawks (kites, harriers, accipiters, and buteos), falcons (including kestrels), ospreys, owls, and vultures. Raptors range in size from the 7-inch little owl to the 47-inch cinereous vulture. Most species have characteristic hunting styles such as soaring (vultures, eagles, and hawks), flying low (harriers), dense forest ambush (accipiters), hovering (kites and kestrels), and watching from perches (buteos and owls). Multiple species of raptor inhabit NASSIG throughout the year (Figure 31). Many migratory species fly from northern Europe down into Africa, passing through Sicily, as they transition between wintering and breeding habitats.



Attractants: Raptors are attracted to habitats that have an abundant supply of prey species. For raptors such as: common buzzard, common kestrel, Eurasian marsh harriers (*Circus aeruginosus*), hen harrier (*Circus cyaneus*), Montagu's harrier (*Circus pygargus*), barn owl, and little owl, prey species include rabbits, small rodents, birds, reptiles, and insects. To hunt effectively, many raptors need large trees or snags (dead standing trees) at the edge of fields in which to perch as they watch for prey. Airports provide several attractants for raptors including perching locations on lights, towers, and radars, as well as open areas to hunt for small mammals. Warm air currents (thermals) reflecting off of the runways and taxiways are also an attractant. Raptors use thermals to soar and glide over open areas while hunting (Gill 1994). Raptors often adapt and use other perches found at airports such as glide slope and localizer antennas, windsocks, or any other structure which they can use to gain an elevated view of their surroundings; this often places them in close association with aircraft operations. The abundance of perches and short grass (<6 inches), coupled with high small mammal numbers (mice and rats), often make airports prime foraging areas for many species of raptor.



Raptors are drawn to the airfield at NASSIG by high populations of European rabbits. European rabbits inhabit many of the grassy areas on the airfield where they breed and feed on a variety of plants. European rabbit populations are concentrated in the approach areas of Runway 10R and may draw raptors to the airfield, increasing BASH risks to aircraft (see 7.4.7 Mammals).

Risks: Raptors represent a significant hazard to aircraft because they are typically large, and their hunting behavior predisposes them to collisions with aircraft. Nationally, raptors account for 15.3% of known species bird-aircraft strikes in the U.S. (Dolbeer et al. 2023). Raptors are most struck on airports while hunting because their focus on prey leaves them vulnerable.



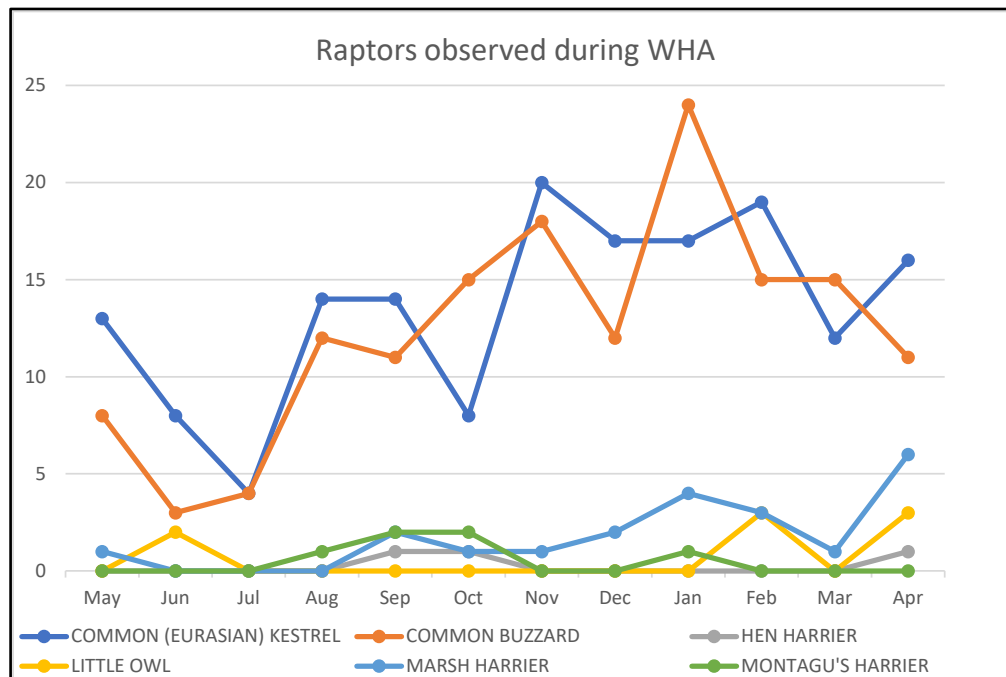
Two common kestrels, one common buzzard, and two Eurasian marsh harriers were known to have been struck by aircraft at NASSIG during the WHA period. Two other common kestrels, two “falcon species,” and one “kite, hawk, eagles, etc. species” were recorded in the ten-year strike record. One of the “falcon species” strikes was a Class D strike, estimated to cause \$31,944 in damage.

Legal status: Raptors in general are protected by L157/92 and other different laws and acts. Common kestrels, barn owls, and little owls are listed in Appendix II of the Bern Convention. The harriers (marsh, hen, and Montagu’s) are listed in Appendix I of the Birds Directive, Appendix III of the Bern Convention and Appendix II of the Bonn Convention. Common buzzards and common kestrels are both listed in Appendix II of the Bonn Convention, and common buzzards are also listed in Appendix III of the Bern Convention.

Control measures: Raptors may be best managed through an integrated program that includes habitat management, rodent and insect control, and harassment. Relocation and lethal reinforcement of harassment techniques would be useful if they are allowed in the future by changes to the FGS and a permitted “derogation” to EU law. The habitat should be managed so that the grass is a uniform composition, density, and height (Cleary et al. 2005). This limits the available prey base of rodents and insects. The trees at the edge of the AOA, which raptors were observed using, should be removed. Spike strips can be installed on structures around the airfield to prevent raptors from perching. If rodent and insect populations increase, rodenticides and insecticides may be applied. If specific raptors habitually frequent an area, they should be routinely dispersed with pyrotechnics.

Prevalence: Raptors were observed year-round on the airfield (Figure 31) with a spike between October and March. This spike likely correlates with migration through the area and the establishment of breeding territories. Numbers then decrease as birds fledge and migrate away.

Figure 31. Raptors recorded during WHA airfield surveys at NASSIG.





7.3.9 Shorebirds

Description: Shorebirds are typically found foraging on invertebrates along the edges of water bodies and areas of temporary standing water where rainwater has collected. These species vary in size from small to medium birds. Shorebirds are attracted to the temporarily flooded areas that are common during heavy winter rains. Shorebirds migrate through the area around NASSIG and are rarely seen outside of the winter months.

Two species of shorebird, the European golden-plover (*Pluvialis apricaria*) and little ringed plover (*Charadrius dubius*), were recorded during WHA airfield surveys at NASSIG. There were additional shorebirds observed that were not identified to species; they were recorded as “unknown shorebird spp.” due to the complexity of species identification combined with a small observation window and large number of birds being counted.



Attractants: Shorebirds are attracted to shallow water (ditches or temporary water) where they can hunt for amphibians and invertebrates. No shorebird nests have been observed on NASSIG but the sparse rocky habitat, especially in the construction area could provide ideal nesting habitat.

Risks: In the U.S., shorebirds represent 9.3% of all known bird-aircraft strikes (Dolbeer et al. 2023). Smaller shorebirds tend to nest close to where they feed, and they forage and travel along shorelines. In comparison, larger shorebird species may travel farther between foraging areas and their roost, potentially crossing various types of terrain, including runways.

Two European golden plovers and one Eurasian stone-curlew (*Burhinus oedipnemos*) were known to have been struck by aircraft at NASSIG during the WHA period. One other Eurasian stone-curlew and one common snipe (*Gallinago gallinago*) were recorded in the ten-year strike record. All shorebird strikes were non-damaging.

Legal status: Each shorebird identified or struck are listed in different sets of laws, but all are listed in Appendix II of the Bonn Convention. European golden plovers are listed under L157/92, Appendix I, II (section 2) and III (Section 2) of the Birds Directive, and Appendix 3 of the Bern Convention. Little ringed plovers are listed under L157/92, and Appendix II of the Bern Convention. Eurasian stone-curlew are listed under L157/92, Appendix I of the Birds Directive and Appendix II of the Bern Convention. Common snipes are listed under Appendix II (section 2) and III (Section 2) of the Birds Directive and Appendix III of the Bern Convention.

Control measures: Airport drainage ditches should be kept free of aquatic vegetation and tall grass. Standing water should be drained from ditches to reduce attractiveness to shorebirds. Hazing with pyrotechnics should occur if these birds are observed foraging, roosting, or loafing on the airport.

Prevalence: Shorebirds were the least recorded species on the airfield at NASSIG during WHA data collection, making up less than 1 percent of all animals recorded (Figures 20 [above]). Total shorebird numbers on the airfield peaked in January during the wet, winter period.



7.3.10 Small Songbirds

Description: This guild includes a wide variety of perching birds that include insectivorous species as well as species attracted to seeds and berries, and species that feed on both. All the species in this group are small sized birds, with observed birds in this guild ranging from the common redstart (*Phoenicurus phoenicurus*) at .21 oz to the Italian sparrow that ranges up to 1.4 oz. Many of small songbirds prefer to forage in the mowed grass commonly found along the edges of taxiways and runways; this makes them a strike risk during certain times of the year. Some species of small songbird are classified as year-round residents, while other species migrate to various areas in Europe and Africa.



Italian sparrow, Photo by L. Petersson

Attractants: Many songbird species are attracted to large, open grassy areas in which they forage for insects, worms and seeds. Long grasses that produce seed are common on and around the airfield. Many species in this group use the large expanses of grassland areas regardless of the mowing regime, while others tend to prefer the shortly mowed grass areas for foraging. Edge habitat, or anywhere where two (or more) habitat types meet, are also common on the airfield and particularly attractive to songbirds.



Goldfinch, Photo by S. Carrera

Risks: In the U.S., this group represents around 11% of all known strikes that occurred between 1990 and 2022 (Dolbeer et al. 2023). The average body size of small songbirds does not make them particularly dangerous to air traffic singularly, but they pose a significantly higher threat when they are grouped into flocks. During migration, songbirds fly in large flocks and often at night to evade predators. Eight songbird strike events were reported to have occurred at NASSIG during WHA data collection (Table 1) that included one Class E damaging strike that had \$540 in damage. Another five non-damaging songbird strikes were listed in the ten-year strike record.

Legal Status: With the vast variety of species observed, struck, or having the potential to appear at NASSIG, the individual status of each species cannot be succinctly listed. Legal status of every species can be found in Appendix 4. In general, all small songbirds are protected by Italian and EU law and thus also protected by the FGS.

Behavior: Songbirds were observed flying, feeding, vocalizing, and loafing. The proportion of vocalizing observations indicate that many Italian sparrows are likely residents at NASSIG and have established territories on the airport. Songbird observations on the airfield included large flocks of Italian sparrow and goldfinches (Figure 33), likely migrating through to their breeding area, indicating a seasonally dependent, slight to high hazard to airport operations. Many songbird will also migrate in large, mixed species flocks at night, resulting in a strike hazard not present throughout the majority of the year.

Control Measures: Songbirds can be effectively managed by habitat modifications, insecticides, and harassment. Some of the habitat modifications, mainly mowing, that could be implemented for managing species in this guild are contrary to FAA/DOD regulations, as well as wildlife management efforts for other species, making this type of management useful in only very specific situations. If large numbers of insects are present in areas where these species are observed, insecticides can be applied to reduce the insect



population, thus decreasing the overall attractiveness of these areas. Harassment can be an effective tool if larger flocks of some species are observed; however, species in this guild are often very persistent and require extended periods of harassment. As with any harassment effort, persistence is the key to success.

Songbirds are typically difficult to manage with pyrotechnics because they only temporarily adjust flight or foraging patterns. Habitat modifications such as vegetation removal, changing mowing regimes, and applying herbicides and insecticides may also aid in reducing the presence of songbirds. Habitat modification recommendations can be found in a latter section of this report (9.2 Habitat Recommendations). Habitat modification will reduce the seeds and cover available to songbirds.

Prevalence: Eighteen species grouped into the songbird guild were observed on the NASSIG airfield during WHA airfield surveys. Of these species, Italian sparrow and European goldfinches (*Carduelis carduelis*), were the most problematic from a BASH viewpoint due to periodic high numbers and behavior (Figure 32-33). These two species made up 87% of all songbirds observed during WHA airfield surveys. Italian sparrows peaked on the airfield in October. This species was periodically concentrated in large groups flying across the airfield and were likely migrating through the area. Goldfinch numbers were relatively high at NASSIG from October through April (Figure 33). This species was commonly found loafing in large flocks in the pastures in the approach of runway 10R.

Figure 32. Small Songbirds recorded during WHA airfield surveys at NASSIG.

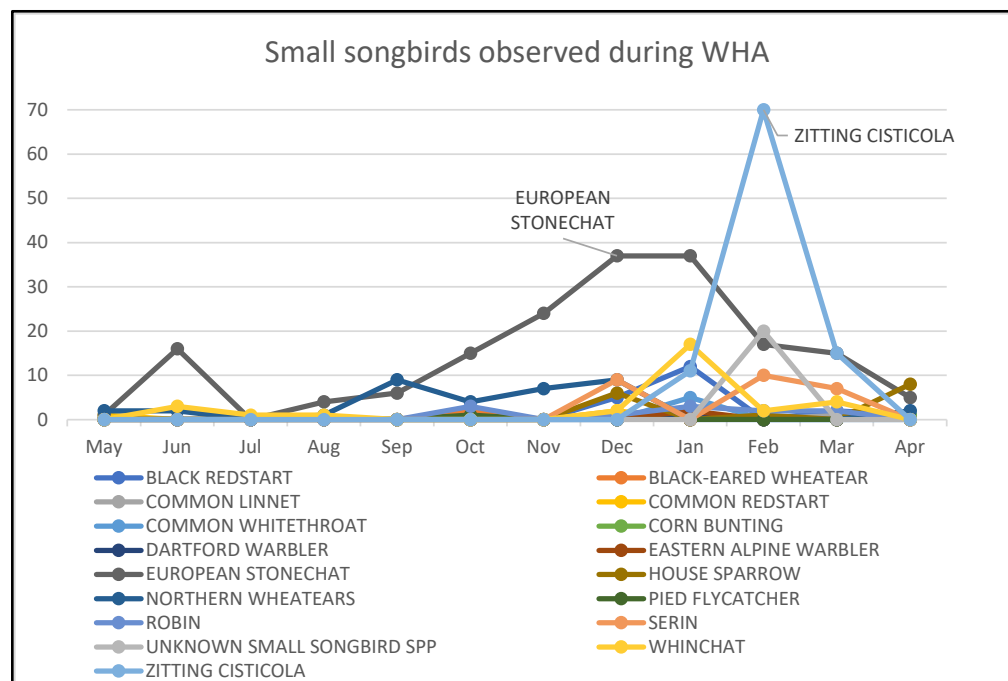
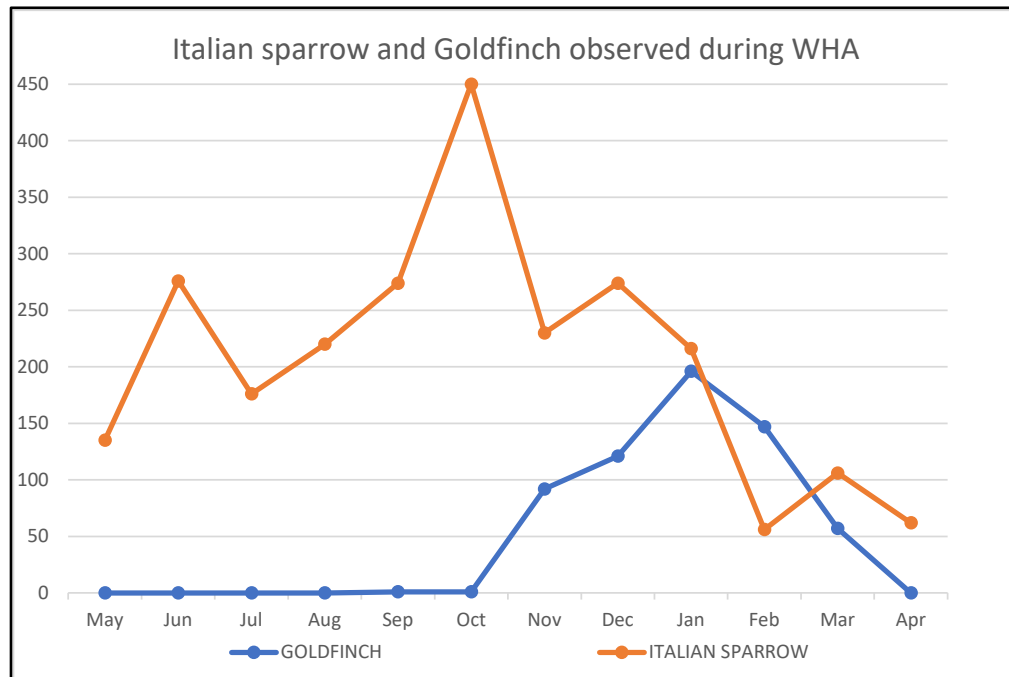


Figure 33. Italian sparrow and European goldfinch recorded during WHA airfield surveys at NASSIG.





7.3.11 Wading birds

Description: The wading birds guild is usually thought of as the “long-legged” and “long-necked” shorebirds guild. Long-legged shorebirds, or wading birds, are comprised mainly of storks, egrets, and herons; the “short-legged” shorebirds including plovers, curlews, etc. are discussed in 7.4.9 Shorebirds. Some of the birds observed in this group at NASSIG include black-crowned night heron (*Nycticorax nycticorax*), great egret (*Ardea alba*), little egrets, western cattle egrets, gray heron, and white stork (*Ciconia ciconia*). These birds use their long legs to wade through shallow water and tall grasses while hunting vertebrate and invertebrate prey species. Their larger size poses an increased risk of damage if they are involved in a bird strike.



Little egrets (left) and gray heron (right),
Photo by R. Mitchell

Attractants: Wading birds are attracted to shallow water (ditches or temporary water), marsh areas, or short grass areas with ponding where they can hunt for small to medium-sized fish, amphibians, small mammals, and insects. Many of these bird species spend most of the time near/in water, but will occasionally forage in grassland areas, depending on available food sources.

Risks: In the U.S., this group represents around 11% of all known strikes that occurred between 1990 and 2022 (Dolbeer et al. 2023). Egrets and herons pose a more serious threat to aircraft than their smaller



White stork, Photo by K. Kulacoglu

relatives. These long-legged wading birds are larger and can cause greater impact damage to aircraft. They are also slower and more lumbering in flight than the smaller birds. The larger birds may travel farther between foraging areas and their roost, potentially crossing various types of terrain, including runways. Mixed flocks of cattle egrets and little egrets were routinely present in the grassy areas on the airfield where they hunt for small mammals and other prey. Other wading bird species, including white storks, were observed foraging in the drainages on the east side of the airfield. Two cattle egrets were struck with no damage at NASSIG during WHA data collection. One white stork was struck on 25APR2022

resulting in a Class B bird strike to a P-8A, destroying an engine and estimated to cost \$913,100. One water rail (*Rallus aquaticus*) was also listed in the ten-year strike record in a non-damaging strike.

Legal status: Wading birds identified or struck are listed under different sets of laws. Black-crowned night herons, great egrets and little egrets are listed in L157/92, Appendix 1 of the Birds Directive, and Appendix 2 of the Bern Convention. Cattle egrets are listed in L157/92, and Appendix 2 of the Bern Convention. Gray herons are listed in L157/92 and Appendix 3 of the Bern Convention. Water rails are listed under Appendix 2, section 2 of the Birds Directive and Appendix 3 of the Bern Convention. Finally, white storks are listed under Article 2 of L157/92, Appendix 1 of the Birds Directive and Appendix 2 of the Bern Convention.

Control measures: Airport drainage ditches should be kept free of vegetation and tall grass on the edges. Standing water should be drained from ditches to reduce attractiveness to wading birds. Egrets, herons, and storks should be deterred from crossing the airport by hazing them with pyrotechnics. Hazing with pyrotechnics should also occur if these birds are observed foraging, roosting, or loafing on the airport.



Prevalence: Six species of wading birds were recorded on and near the airfield during WHA airfield surveys. Wading bird numbers peaked on the NASSIG airfield in September and October (Figure 34-35). Cattle egrets were the most observed species, making up 84% of all wading birds recorded. Little egrets (10%) and gray herons (5%) followed in total numbers recorded. Looking at seasonal trends in wading bird numbers, cattle egrets peaked in September and October and then again in April (Figure 35). Little egrets peaked in March. Gray herons are present year-round but peaked in October and again in January-February which is likely due to migration. Other species were observed throughout the year in low numbers (Figure 34).

Figure 34. Wading birds recorded during WHA airfield surveys at NASSIG (excluding cattle and little egrets).

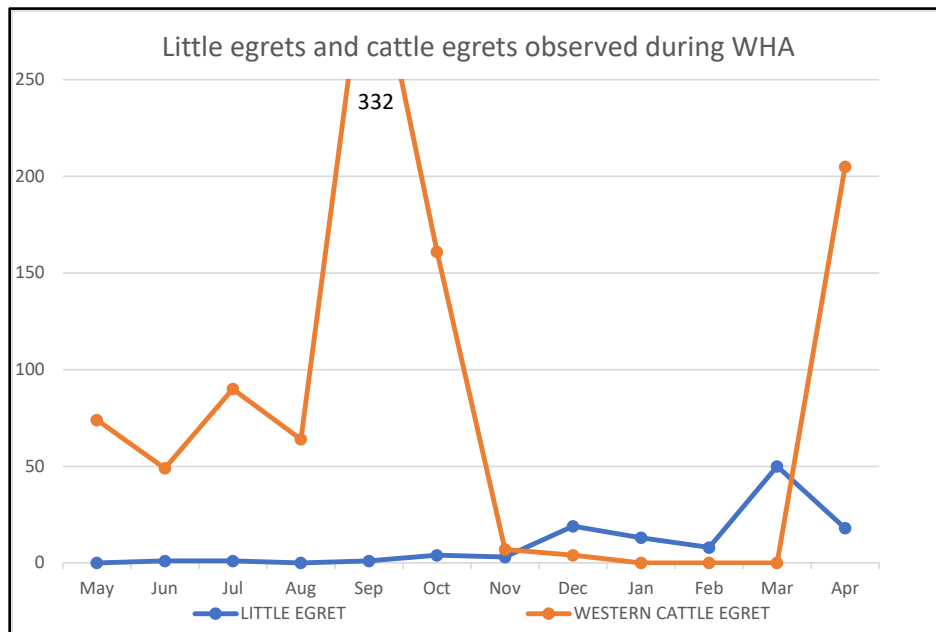
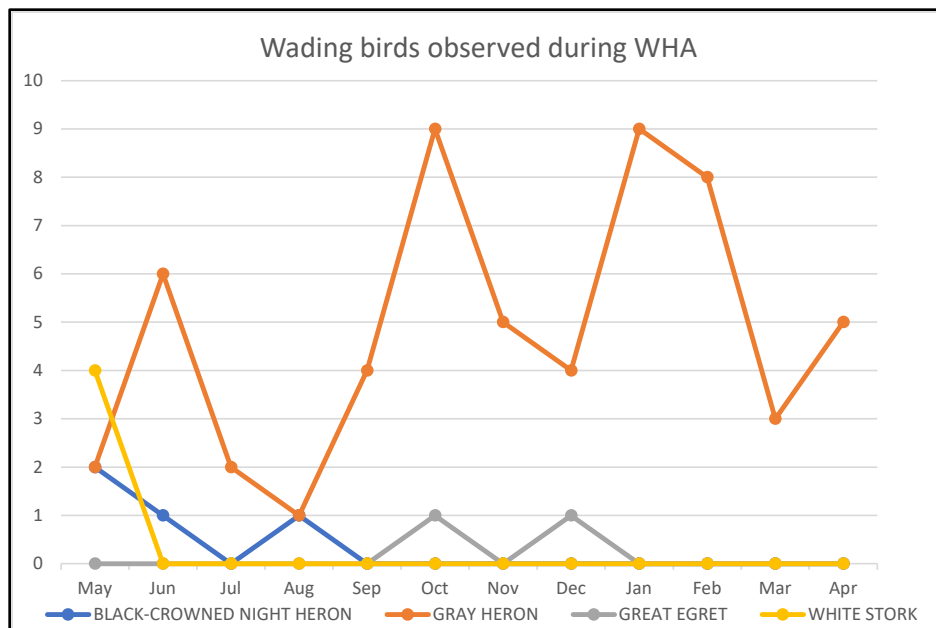


Figure 35. Cattle egrets and little egrets recorded during WHA airfield surveys at NASSIG.





8.0 CONCLUSIONS

The main objective of any BASH program is to reduce the risk of a wildlife/aircraft strike. USDA-WS' job is to identify a hazard, mitigate the hazard, and make recommendations to reduce that hazard and the possibility of other hazards from appearing in the future, to the greatest extent possible. There will always be a need for the BASH Program at NASSIG because the active runways and movement areas are encircled by hazardous habitats and the continuous evolving nature of the base. The ever-present risk of wildlife related damage to aircraft and the potential for injury/death of aviators reinforces the purpose and justifies the need for a BASH program at NASSIG.

9.0 RECOMMENDATIONS

The following recommendations are intended to assist in improving the wildlife hazard management program at NASSIG and aid in reducing and/or eliminating wildlife hazards documented during this WHA. They are recommendations, not mandates, and are to be implemented at the airfield's discretion, with specific hazards/goals being prioritized and mitigation completed when financial and/or labor resources are available. USDA-WS recognizes some of the recommendations are simple and require little planning. However, other recommendations involve the coordination of people from several different entities including NATO and ITAF, require various resources and funding, and are intended to be long-term projects.

Recommendations herein replace NASSIG's Wildlife Hazard Management Plan (WHMP) IAW CNICINST 3750.1B ONCONUS requirements and will result in a reduction of wildlife hazards to aircraft operating at NASSIG. However, continuation of a monitoring program to assess wildlife activity is necessary to identify new hazards that may arise daily, seasonally, annually, or due to other changes such as landscape modifications, weather patterns, and environmental changes. In addition, wildlife management efforts need to be monitored, documented, and if need be, modified to ensure effectiveness and completion of program goals and objectives. BASH management will be prioritized based on the defined BASH Mitigation Zone (BMZ) (Appendix 5).

9.1 General Recommendations

1. USDA-WS recommends airfield users heed USDA-WS and Wildlife Detection and Dispersal Team (WDDT) wildlife notifications (including BWC recommendations) relayed directly to the tower regarding wildlife activity on the airfield and potential flight risk areas. Ultimately, the BWC of the airfield is determined by ITAF ATC, but U.S. aircrews should take notice of Automatic Terminal Information Systems (ATIS) messages and published Notice to Airmen (NOTAMS) that contain bird condition information that may contradict the current airfield BWC.
2. Due to multiple ongoing construction projects, the airfield environment is experiencing dramatic changes. The extension of both runways by over 3000 feet and the ITAF eminent domain seizure of the land to allow for it, will cause the footprint of the airfield to grow into new habitats. There are also plans to build aircraft parking ramps on the south side of the airfield near the fuel facility. Although USN airfields are not controlled by FAA requirements, CFR, Title 14, Part 139.137 would require a new WHA be completed to adequately capture the "current" airfield environment and wildlife hazards after such major changes. Continual monitoring may be able to capture the changes in real time, but not with the depth of analysis that a full WHA would accomplish.



3. USDA-WS recommends that NASSIG continue to enforce the wildlife strike reporting procedure. Pilots, airport operators, aircraft maintenance crews, and ground crews should be instructed by the NASSIG ASO to complete a NATO Wildlife/Aircraft Strike Remains Reporting Form each time there is a wildlife strike and enter the strike information into RMI or the Air Force Safety Automated System (AFSAS), as appropriate. All strikes should also be recorded by the ASO in the NASSIG bird strike tracker. Bird strike reports should include the aircraft type, date and time, location, aircraft damage, phase of flight, bird species, bird remains, monetary loss if available, and a description of the bird strike (Appendices 1-2). The report should be completed immediately after the strike. Collecting this information and data can help prevent future wildlife strikes and maintain accurate records of strikes at NASSIG. Reporting reminders should be given on a consistent basis through meetings, presentations, and/or personal communications. USDA-WS recommends wildlife strike reporting instruction be included in new staff briefings/materials. Bird/wildlife strike kits should be maintained by each squadron to have ready. Strike kits can be requested from USDA-WS as needed.
4. Continue and maintain the BWG Quarterly Meetings - Per CNIC Instruction 3750.1C and the NASSIGINST 3750.3F requirements, airfields shall establish a BWG to deal with wildlife related issues on the airfield environment. Membership shall consist of all parties involved with Air Operations, Tenant Squadrons, Natural Resources, ITAF and/or other programs affected by BASH issues and will be overseen principally by the Air Operation Officer (AOPS) at NASSIG. This group will be capable of sharing pertinent information that can be used to recommend actions and changes in operations to reduce hazards on the airfield. The BWG shall meet quarterly to discuss all wildlife issues posing a potential risk to aviation at NASSIG and proposed ideas for dealing with these issues and progress reports on previous work.
5. Continued wildlife data collection at NASSIG is recommended. Documenting wildlife activity is crucial to interpreting and illustrating wildlife hazards to aircraft. Wildlife activity can be easily organized using data sheets incorporated into a computer database (i.e., Microsoft Excel) or entered into mapping software and represented both spatially and in database form (i.e., ArcGIS), which will then allow these records to be presented as charts, graphs, and reports. Comprehensive data collection can also provide immediate and accurate reporting to justify management actions. Furthermore, organized record keeping can be beneficial to airport personnel during investigations in the aftermath of a damaging wildlife strike. USDA-WS can continue using the standardized data recording system that mirrors the methodology used in this WHA.
6. Update FGS Italy and research EU and Italy Wildlife Law Derogations - The ability to respond to wildlife in a prompt and efficient manner is essential for providing safe aviation operations at NASSIG. Lethal control of wildlife and removal of active bird nests is sometimes necessary to deal with hazardous wildlife that pose an immediate threat to air operations/safety either through large numbers or specific behaviors that increase the risk of bird strikes. The current FGS does not include language that would allow the use of “derogations” to current wildlife laws. If the FGS can be updated to allow for derogations, USDA-WS would be able to move forward with other control methods including trapping and relocation of raptors, trapping and lethal removal, and potentially other methods. Most methods can be completed without the need of firearms. Updates to FGS Italy will be completed at the CNREURAFCENT level or above; however, NASSIG can champion the effort and increase the chances of success.



7. Ensure all airfield users receive annual BASH refresher training as required by CNICINST 3750.1B. Due to the varied rotations of personnel at NASSIG, adding the BASH training to the Course Rules Briefs and Airfield Vehicle Operators Course (AVOC) training may be the best option to ensure the annual requirement is met. Records of personnel trained must be maintained locally.
8. Recent efforts have been made to prepare Navy personnel for dispersing wildlife from the airfield in the form of a WDDT. USDA-WS recommends these actions continue. Personnel should be available to alleviate immediate BASH problems should they arise when USDA-WS personnel are not available to assist. CNIC Instruction 3750.1B describes procedures for forming a WDDT. Team members' responsibilities involve harassing any wildlife that are posing a direct threat to air operations on the airfield. Each member should be trained and familiarized with air operations, the airfield environment, and wildlife harassment/mitigation procedures, which should be renewed annually. Additionally, each member shall complete the AVOC course given by Air Operations to hold an airfield driver's license that must be renewed annually. All work shall be coordinated and communicated to the Air Traffic Control tower through direct communications and recorded with pertinent information such as date, species, and type of dispersal device (i.e., vehicle). These records should be maintained and compiled for monthly and annual reporting purposes to the BWG. WDDT Level 2 includes training for the safe use of pyrotechnics and can be overseen by USDA-WS. Wildlife dispersal/control supplies should only be used by approved, trained personnel, and pyrotechnics should be stored in an approved Ready Service Locker (RSL) when not in use. At a minimum, trained and approved personnel who operate regularly on the airfield should have been provided a BASH "go-bag" and have the following equipment available:
 - Radio for communication with tower
 - Binoculars
 - Bird and mammal identification guide
 - Data collection materials such as daily logs that can be used to enter wildlife control data for presentation at BASH WG meetings.
 - **LEVEL 2 Trained Personnel**
 - 15 mm pyrotechnic pistol launcher and caps
 - Bird Bangers
 - Screamer Sirens
 - An approved RSL for pyrotechnics storage
 - Safety equipment (ear and eye protection)
9. USDA-WS recommends that NASSIG research and potentially install a bird detection radar system. Currently, NASSIG ATC staff and pilots rely on the daily avian reports from airfield patrols to determine BWCs. Though this is approved per installation BASH guidelines and has shown to be successful in reducing aircraft strikes, a real-time avian detection radar system would allow for immediate BASH condition relay. Some systems can be programed to update ATIS directly, which would allow for pilots to know the exact location of bird activity on and around the airfield. Currently, there are two major systems being used at various DOD airfield installations, Accipiter and Merlin. Depending on which system, version, complexity, and scope would be installed at NASSIG, price could range from \$125,000 to \$600,000+ for installation with an additional \$2,500 to \$10,000 per year for maintenance costs.



9.2 Habitat Recommendations

Managing airfield habitats is considered the most important approach for reducing wildlife use at airports (Cleary and Dolbeer 2005, Washburn and Seamans 2007). Vegetation management, including airfield mowing and vegetation removal, is the most pressing habitat project needed at NASSIG. Addressing vegetation on the airfield will likely have the most drastic effect on wildlife. Drainage improvement and removal of obsolete structures from within the PSA are also important. The CNIC BASH Manual has guidelines and information on habitat manual, while the CNICINST 3750.1B contain guidance on responsibilities in planning projects, and prioritization of projects and funding.

USDA-WS recommends that all airfield grass be mowed IAW the ITAF WASH Plan which allows for a max height of 40cm (15.75 inches) and shorter heights closer to the runways (Figure 11, above) (with awareness of keeping edge habitat further from the active surfaces). Mowing should occur before herbaceous plants on the airfield can flower and seed. Ideally, the mowed areas can be expanded to include all grass within the perimeter fence around the airfield or at a minimum, within the inner perimeter road. There are many mounds and uneven areas that will need to be leveled to allow for easier maintenance with the existing tractor. Areas that cannot be leveled will need to be maintained using hand equipment to eliminate edge effects between different grass heights. Disking of any area on the airfield should be discouraged. Disking creates areas frequently used by flocks of pigeons for grit".

1. USDA-WS recommends removal of all trees on the airfield (Figure 12, above), especially the two along the runway near Ramp 6. These trees are utilized by numerous large-bodied birds, mainly common buzzards, magpies, and common kestrels. Many are utilized by groups of Italian sparrows and common starlings as well. Tree can also constitute a "clear zone" violation by obstructing flight paths and safety zones, providing additional (if not the primary) rationale for removal.
2. All drainages on the airfield should be improved and maintained. The drainage flowing from the 28L Approach should be the main priority. Drainages should flow freely and not have any standing water that can attract wildlife. Vegetation along drainages should be mowed to the bank, with vegetation in the drainage permitted to grow as described in the 2017 NASSIG WHA. Drainages can also be put into an underground, enclosed culvert, when possible, to decrease yearly vegetation maintenance.
3. Lights were installed below the existing bird netting in Hangar 630 and 633 for ease of installation. This has caused the light fixtures to be easily accessible to rock pigeons that loaf and roost on them. Although some light fixtures have bird spikes installed, they are not in a high enough density to exclude the pigeons. The spikes are also not attached in a way that ensures they will not become unattached and pose a FOD hazard. Lights should be raised above the existing bird netting and bird netting should be monitored for holes. Any holes created for hangar or systems maintenance must be repaired upon completion of the project.
4. All structures, light poles, airfield markers, and signs should be evaluated for operational necessity. Any unnecessary objects should be scheduled for removal by ITAF. All necessary objects near the airfield should be fitted with an appropriate anti-perching device or exclusionary device (e.g. Nixalite®, Birdwire®, Daddi Long Legs™, grid wires, netting, etc.) to discourage birds from utilizing these platforms. These materials should be installed and regularly maintained. Priority should be given to objects within the red area of Figure 36. Unused buildings (Figure 13, above) within the priority area should be removed.



Figure 36. Perch Removal Priority Area



9.3 Wildlife Management Recommendations

1. Maintain management efforts to reduce, discourage, and eliminate the number of birds and mammals that nest, roost, forage, and loaf in areas around the airfields, thereby reducing the strike hazard to aircraft. This includes use of pyrotechnics and vehicle harassment. Early initiation and continuous application of the BASH program is the key to its success.
2. European rabbit numbers near the airfield should continue to be monitored. Collaboration with ITAF and the Italian *Forestate* may allow for more efficient rabbit management on the airfield
3. USDA-WS recommends that the perimeter fence be repaired or replaced, prioritizing the areas near the airfield. The fence begins at ground level. Feral dogs can burrow under the perimeter fence, making their way onto the airfield and across runways and taxiways, causing a BASH risk. The fence should be extended into the ground as illustrated in Appendix 6 to eliminate mammal crossings. The concrete along base of the fence could also be reinforced with additional concrete or a thick layer of gravel running perpendicular to the base of the fence.
4. Periodic large insect hatches on the airfield attracted large groups of various swallow species, as well as other bird species, to the airfield during the WHA. USDA-WS recommends insecticide applications to the airfield to reduce this BASH risk as long as the insecticide can be applied safely with regard to personnel operating on the airfield.



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Appendix 1:

NASSIG Strike Remains Collection Procedures.

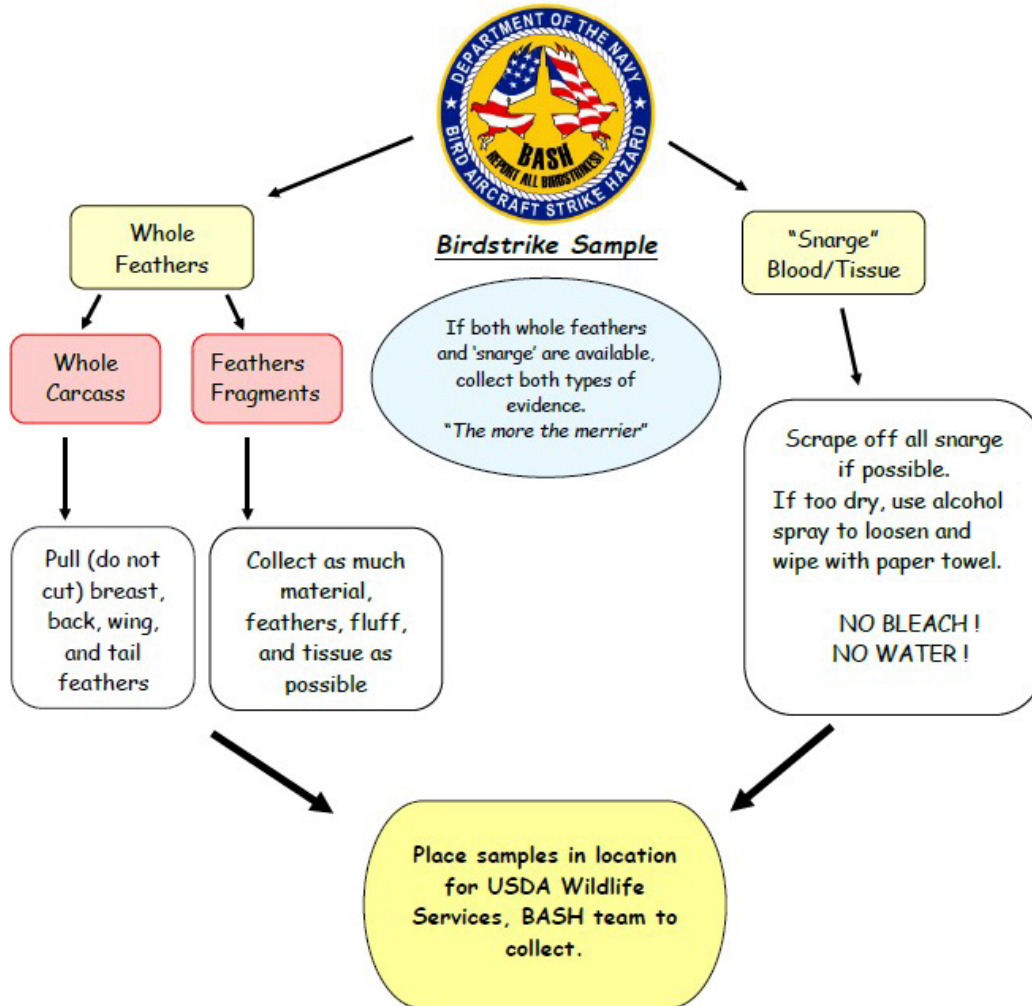
- 1) Record all required information on the report sheet (attached).
- 2) Wear gloves to avoid potential bacteria or virus transmission and avoid DNA contamination.
- 2) Each strike location should be sampled and placed in a separate, labeled small bag. These small bags can be placed in the larger bag along with any unused supplies for future use.
- 3) Sample Snarge using alcohol swab in kit. Feathers located in the snarge sample can be placed in the same bag as the alcohol swab.
- 4) Full carcasses can be placed within the (labelled) 1 gallon bag or a larger plastic bag if necessary.
- 5) Place samples in the fridge in Hangar 630, please fill out the inventory sheet on the door.
- 6) Report strikes as required.



Appendix 1 (continued):

NASSIG Strike Remains Collection Procedures.

BIRD AND BAT REMAINS COLLECTION PROCEDURES





Appendix 2:

NASSIG/NATO Strike Report Form Page 1.

NATO WILDLIFE STRIKE FORM			
(Complete and deliver to the Flight Safety Office within 24 hours)			
REMEMBER			
SAVE WITH OTHER NAME IN ORDER TO MAINTAIN THE ORIGINAL MODEL ALWAYS AVAILABLE			
01. Adm number		10. Flight phase	☐ Parked
02. Incident date			☐ Taxing
03. Incident time (local)			☐ Take off Roll
04. Unit Squadron			☐ climb
05. Aircraft type e Reg nr.			☐ Low level operations
06. Speed			☐ Cruise
07. Altitude //			☐ Holding in circuit
08. Lat. Long. Coord. //			☐ Descent
09. Geogr. a.. Country ITALY			☐ Final approach
Location			☐ Landing roll
b. Location SIGONELLA AIRBASE			☐ Other
c. Unknown ☐			☐ Unknown
11. Operational effects	☐ No	☐ Yes	☐ Abort take - off
			☐ Return to base
			☐ Landed at nearest airbase, being
			☐ Other
12. Crew injury	☐ No	☐ Yes	
13. Damage to aircraft	☐ No	☐ Yes	☐ Unknown
14. Strike notice by	☐ Pilot	☐ Ground crew	☐ Other
15. Impact point	16. Damage point	15. Impact point	16. Damage Point
Radome ☐	☐	Canopy Wind screen ☐	☐
Nose ☐	☐	Fuselage ☐	☐
Wing ☐	☐	Landing Gear ☐	☐
Rotor ☐	☐	Empenage ☐	☐
Propeller ☐	☐	Underwing Stores ☐	☐
Air intake ☐	☐	Unknown ☐	☐
Engine ☐	☐		

**Appendix 2 (continued):****NASSIG/NATO Strike Report Form Page 2.**

17.	Remains collected	☐ yes	☐ No	☐ No remains found
18.	Wildlife identified	☐ Laboratory	☐ BCU	☐ Other
19.	Wildlife species	UNK		
20.	Wildlife number	seen	struck	
		☐ Single	☐ Single	
		☐ Flock nr. ()	☐ Flock nr. ()	
		☐ Unknown	☐ Unknown	
21.	In case of bird	☒ Small (sparrow size)	☐ Medium (pigeon)	☐ Large (goose)
		☐ Unknown		
<u>ADDITIONAL INFORMATION</u>				
22.	Pilots name	Fare clic qui per immettere testo.		
23.	Engine reg. Number	//		
24.	Costs:	//		
25.	Man hours spent to fix	//		
26.	Classification (if applicable)	//		
27.	A/C Lights	☐ On	☐ Off	☐ Unknown
28.	BIRDTAM	☐ Yes, level:....	☐ No	☐ Unknown
29.	Local Bird Intensity	☐ Low	☐ Moderate	☐ Severe
30.	Light conditions	☐ Dawn	☒ Day	☐ Dusk
		☐ Night	☐ Unknown	
31.	Conditions	☐ IMC	☐ VMC	☐ Snow
		☐ Rain	☐ Clouds..../8	☐ Visibility
32.	POC / Form filled in by			
33.	Remarks / Comments :			



Appendix 3:

Species Observed at NASSIG During the WHA Period.

Species Observed	Scientific Name	Guild
Barn swallow	<i>Hirundo rustica</i>	Aerialists
Common house martin	<i>Delichon urbicum</i>	Aerialists
Common swift	<i>Apus apus</i>	Aerialists
Common starling	<i>Sturnis vulgaris</i>	Blackbirds
Spotless starling	<i>Sturnis unicolor</i>	Blackbirds
Eurasian collared dove	<i>Streptopelia decaoto</i>	Columbids
European turtle dove	<i>Streptopelia turtur</i>	Columbids
Rock pigeon	<i>Columba livia</i>	Columbids
Wood pigeon	<i>Columba palumbus</i>	Columbids
Common magpie	<i>Pica pica</i>	Corvids
Common kingfisher	<i>Alcedo atthis</i>	Forest birds
Common hoopoe	<i>Upupa epops</i>	Forest birds
Crested lark	<i>Galerida cristata</i>	Grassland birds
Gray wagtail	<i>Motacilla cinerea</i>	Grassland birds
White wagtail	<i>Motacilla alba</i>	Grassland birds
Wood lark	<i>Lullula arborea</i>	Grassland birds
Woodchat shrike	<i>Lanius senator</i>	Grassland birds
Yellow-legged gull	<i>Larus michahellis</i>	Gulls
Brown (Norway) rat	<i>Rattus norvegicus</i>	Mammals
Feral cat	<i>Felis catus</i>	Mammals
Feral dog	<i>Canis familiaris</i>	Mammals
Red fox	<i>Vulpes vulpes</i>	Mammals
European rabbit	<i>Oryctolagus cuniculus</i>	Mammals
Barn owl	<i>Tyto alba</i>	Raptors
Common kestrel	<i>Falco tinnunculus</i>	Raptors
Common buzzard	<i>Buteo buteo</i>	Raptors
Hen harrier	<i>Circus cyaneus</i>	Raptors
Little owl	<i>Athene noctua</i>	Raptors
Montagu's harrier	<i>Circus pygargus</i>	Raptors
Western marsh harrier	<i>Circus aeruginosus</i>	Raptors
Common snipe	<i>Gallinago gallinago</i>	Shorebirds
European golden plover	<i>Pluvialis apricaria</i>	Shorebirds
Little ringed plover	<i>Charadrius dubius</i>	Shorebirds
Black redstart	<i>Phoenicurus ochruros</i>	Small songbirds
Black-eared wheatear	<i>Oenanthe hispanica</i>	Small songbirds
Common linnet	<i>Linaria cannabina</i>	Small songbirds



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Common redstart	<i>Phoenicurus phoenicurus</i>	Small songbirds
Common whitethroat	<i>Sylvia communis</i>	Small songbirds
Corn bunting	<i>Emberiza calandra</i>	Small songbirds
Dartford warbler	<i>Sylvia undata</i>	Small songbirds
Eastern alpine warbler	<i>Curruca cantillans</i>	Small songbirds
European stonechat	<i>Saxicola rubicola</i>	Small songbirds
Goldfinch	<i>Carduelis carduelis</i>	Small songbirds
House sparrow	<i>Passer domesticus</i>	Small songbirds
Italian sparrow	<i>Passer italiae</i>	Small songbirds
Northern wheatear	<i>Oenanthe oenanthe</i>	Small songbirds
Pied flycatcher	<i>Ficedula hypoleuca</i>	Small songbirds
Robin	<i>Erithacus rubecula</i>	Small songbirds
Serin	<i>Serinus serinus</i>	Small songbirds
Whinchat	<i>Saxicola rubetra</i>	Small songbirds
Zitting Cisticola	<i>Cisticola juncidis</i>	Small songbirds
Macqueen's Bustard	<i>Chlamydotis macqueenii</i>	Upland birds
Black-crowned night heron	<i>Nycticorax nycticorax</i>	Wading birds
Gray heron	<i>Ardea cinerea</i>	Wading birds
Great egret	<i>Ardea alba</i>	Wading birds
Little egret	<i>Egretta garzetta</i>	Wading birds
Western cattle egret	<i>Bubulcus ibis</i>	Wading birds
White stork	<i>Ciconia ciconia</i>	Wading birds
Common moorhen	<i>Gallinula chloropus</i>	Waterbirds
Great cormorant	<i>Phalacrocorax carbo</i>	Waterbirds



Appendix 4:

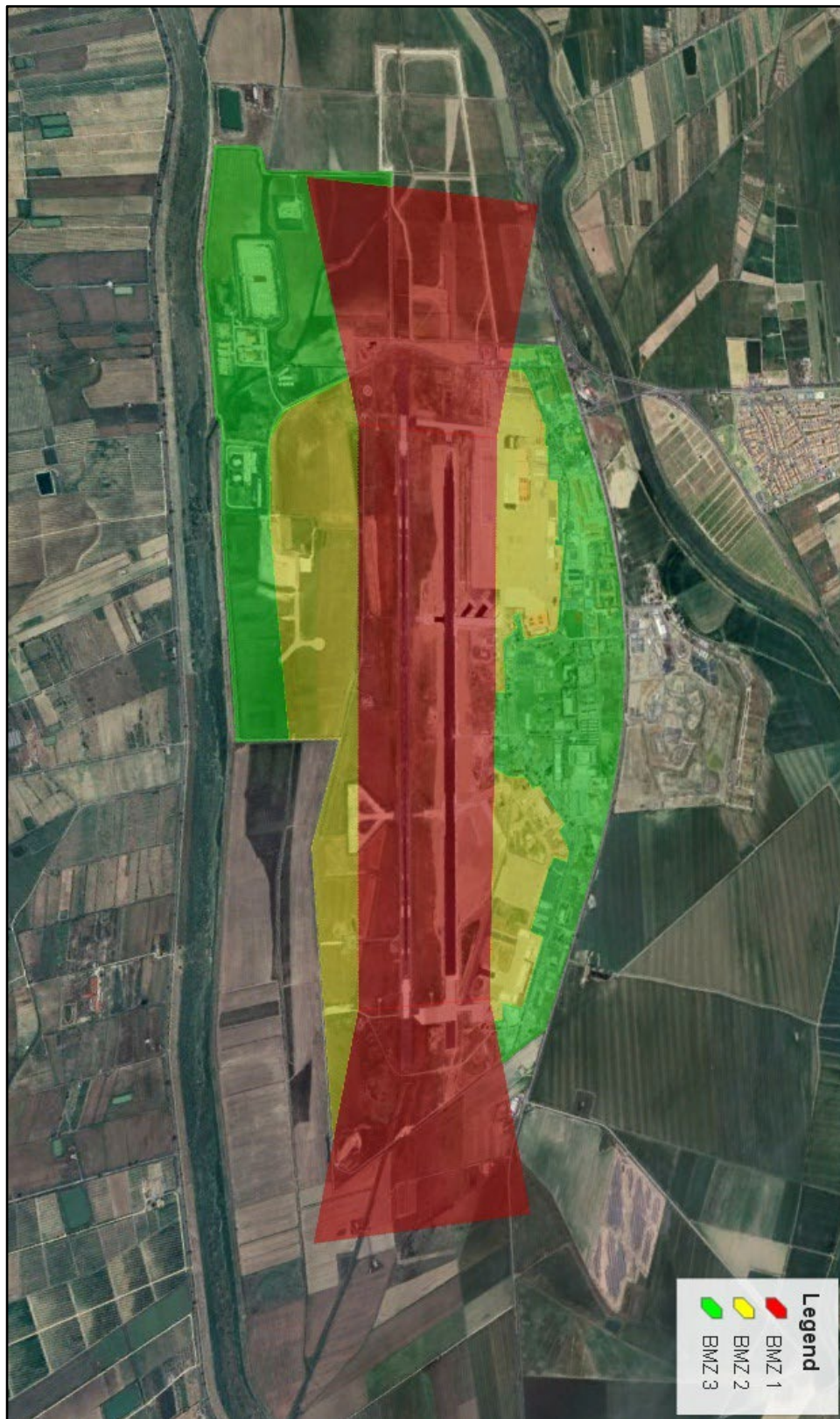
Legal Status of Species Observed at NASSIG During the WHA Period.

classe	ordine	famiglia	specie_lat	specie_it	L. 157/92 art. 2	L. 157/92	79/409 CEE Ap.1	79/409 CEE Ap.21	79/409 CEE Ap.211	79/409 CEE Ap.31	79/409 CEE Ap.311	BERNA Ap.2	BERNA Ap.3	CITES AII. A	CITES AII. B	CITES AII. D	BONN Ap.1	BONN Ap.2	HABITAT Ap.2	HABITAT Ap.4	HABITAT Ap.5	BARCELONA aII. 2	ENDEMICA	CHECKLIST	IUCN
AVES	ACCIPITRIFORMES	Accipitridae	Buteo buteo (Linnaeus, 1758)	Poiana	X																				
AVES	ACCIPITRIFORMES	Accipitridae	Circus aeruginosus (Linnaeus, 1758)	Falco di palude	X																				
AVES	ACCIPITRIFORMES	Accipitridae	Circus cyaneus (Linnaeus, 1758)	Alibonca reale	X																				
AVES	ACCIPITRIFORMES	Accipitridae	Circus pygmaeus (Linnaeus, 1758)	Alibonca minore	X																				
AVES	ACCIPITRIFORMES	Accipitridae	Accipiter nisus (Linnaeus, 1758)	Rondone	X																				
AVES	CHARADRIIFORMES	Charadriidae	Charadrius dubius Scopoli, 1786	Corriere piccolo	X																				
AVES	CHARADRIIFORMES	Charadriidae	Pluvialis apricaria (Linnaeus, 1758)	Piviere dorato	X																				
AVES	CHARADRIIFORMES	Charadriidae	Larus argentatus Pontopidan, 1763	Gabbiano reale nordico	X																				
AVES	CHARADRIIFORMES	Laridae	Larus californicus Pallas, 1811	Gabbiano reale	X																				
AVES	CHARADRIIFORMES	Scotopodidae	Gallinago gallinago (Linnaeus, 1758)	Beccaccione	X																				
AVES	CICONIIFORMES	Ardeidae	Ardea cinerea Linnaeus, 1758	Alcione cinerina	X																				
AVES	CICONIIFORMES	Ardeidae	Bubulcus ibis (Linnaeus, 1758)	Alcione guardabuchi	X																				
AVES	CICONIIFORMES	Ardeidae	Egretta alba (Linnaeus, 1758)	Alcione bianco maggiore	X																				
AVES	CICONIIFORMES	Ardeidae	Egretta garzetta (Linnaeus, 1760)	Garzetta	X																				
AVES	CICONIIFORMES	Ardeidae	Nycticorax nycticorax (Linnaeus, 1758)	Nittora	X																				
AVES	COLUMBIFORMES	Columbidae	Columba oenas (Linnaeus, 1758)	Cicogna bianca	X																				
AVES	COLUMBIFORMES	Columbidae	Columba livia Gmelin, 1789	Piccione selvatico	X																				
AVES	COLUMBIFORMES	Columbidae	Columba palumbus Linnaeus, 1758	Colombaccio	X																				
AVES	COLUMBIFORMES	Columbidae	Streptopelia turtur (Linnaeus, 1758)	Tortora	X																				
AVES	CORACIIFORMES	Alcedinidae	Alcedo atthis (Linnaeus, 1758)	Martín pescatore	X																				
AVES	CORACIIFORMES	Coraciidae	Coracias garrulus Linnaeus, 1758	Gruccone	X																				
AVES	CORACIIFORMES	Meropidae	Merops apiaster Linnaeus, 1758	Gruccone egiziano	X																				
AVES	CORACIIFORMES	Meropidae	Merops superciliosus Linnaeus, 1766	Upupa	X																				
AVES	FALCONIFORMES	Upopidae	Upupa epops Linnaeus, 1758	Upupa	X																				
AVES	FALCONIFORMES	Falconidae	Falco peregrinus Tunstall, 1771	Pellegrino	X																				
AVES	FALCONIFORMES	Falconidae	Falco tinnunculus Linnaeus, 1758	Gheppio	X																				
AVES	GRUIFORMES	Rallidae	Gallinula chloropus (Linnaeus, 1758)	Gallinella d'acqua	X																				
AVES	PASERIFORMES	Alaudidae	Galerida cristata (Linnaeus, 1758)	Cappellaccia	X																				
AVES	PASERIFORMES	Alaudidae	Lullula arborea (Linnaeus, 1758)	Totavilla	X																				
AVES	PASERIFORMES	Fringillidae	Carduelis carduelis (Linnaeus, 1758)	Fanello	X																				
AVES	PASERIFORMES	Fringillidae	Carduelis arvensis (Linnaeus, 1758)	Cardellino	X																				
AVES	PASERIFORMES	Fringillidae	Serinus serinus (Linnaeus, 1758)	Verzellino	X																				
AVES	PASERIFORMES	Hiunderidae	Delichon urbica (Linnaeus, 1758)	Balestruccio	X																				
AVES	PASERIFORMES	Hiunderidae	Hirundo rustica Linnaeus, 1758	Rondine	X																				
AVES	PASERIFORMES	Laniidae	Lanius excubitor Linnaeus, 1758	Aquila caprossa	X																				
AVES	PASERIFORMES	Monticolidae	Monticola alba Linnaeus, 1758	Belluana bianca	X																				
AVES	PASERIFORMES	Monticolidae	Monticola cinerea Tunstall, 1771	Belluana gialla	X																				
AVES	PASERIFORMES	Passeridae	Ficedula hypoleuca Pallas, 1764	Bella nera	X																				
AVES	PASERIFORMES	Passeridae	Passer domesticus (Linnaeus, 1758)	Passera oltremontana	X																				
AVES	PASERIFORMES	Passeridae	Passer hispaniolensis (Tremmink, 1820)	Passera sarda	X																				
AVES	PASERIFORMES	Sturnidae	Sturnus vulgaris Linnaeus, 1758	Sturno	X																				
AVES	PASERIFORMES	Sylviidae	Acrocephalus arundinaceus Linnaeus, 1758	Camareccione	X																				
AVES	PASERIFORMES	Sylviidae	Cisticola juncidis (Rafinesque, 1810)	Beccamoschino	X																				
AVES	PASERIFORMES	Sylviidae	Sylvia cantillans Pallas, 1784	Sierpazzolina	X																				
AVES	PASERIFORMES	Sylviidae	Sylvia communis Latham, 1787	Sierpazzolina	X																				
AVES	PASERIFORMES	Sylviidae	Sylvia undata (Bodder, 1783)	Magliana	X																				
AVES	PASERIFORMES	Turdidae	Erithacus rubecula (Linnaeus, 1758)	Pedrosso	X																				
AVES	PASERIFORMES	Turdidae	Oenanthe isabellina Linnaeus, 1758	Monachella	X																				
AVES	PASERIFORMES	Turdidae	Oenanthe isabellina Linnaeus, 1758	Culbianco	X																				
AVES	PASERIFORMES	Turdidae	Phoenicurus phoenicurus Linnaeus, 1758	Codifosso spazzacchino	X																				
AVES	PASERIFORMES	Turdidae	Phoenicurus phoenicurus Linnaeus, 1758	Silvino	X																				
AVES	PASERIFORMES	Turdidae	Saxicola torquata Linnaeus, 1758	Silvino	X																				
AVES	PELECANIFORMES	Trogonidae	Sitta carolinensis Linnaeus, 1758	Silvino	X																				
AVES	STRIGIFORMES	Strigidae	Nyctalus noctula (Scopoli, 1769)	Civetta	X																				
AVES	STRIGIFORMES	Tyrionidae	Tyto alba (Scopoli, 1769)	Burattino	X																				
AVES	STRIGIFORMES	Vesperilionidae	Hypsiglena savi (Bonaparte, 1837)	Pipistrello di Savi	X																				
AVES	STRIGIFORMES	Vesperilionidae	Pipistrellus kuhlii (Kuhl, 1817)	Pipistrello albicollato	X																				
MAMMALIA	CHIROPTERA	Vesperilionidae																							



Appendix 5:

NASSIG BASH Management Zones (BMZ) 1-3

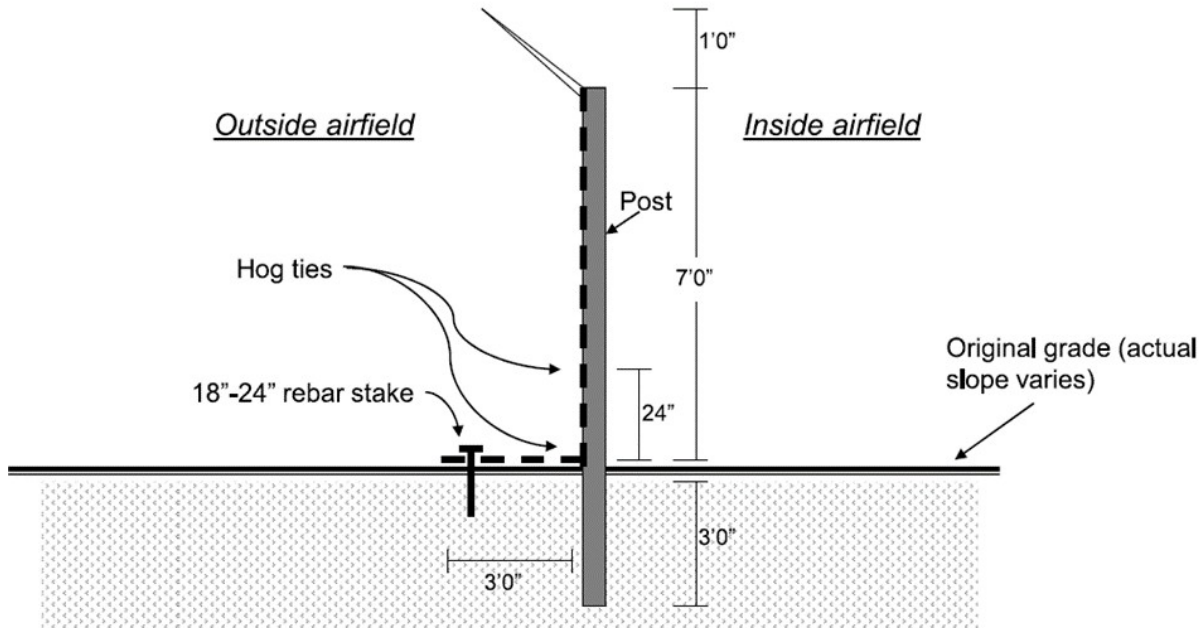




Appendix 6:

Recommended NASSIG Perimeter Fence Replacement Diagram.

Security Fence with Wildlife Deterrent Apron



- Chain link fabric shall either be 10' with 3' loose material staked to ground or 7' with additional 5' attached to bottom portion of fence (2' hogtied to existing fence; 3' staked to the ground)